

**“AUTO TEMPERATURE DETECTOR FOR
ENTRANCE FOR COVID SAFETY”**

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1. ABSTRACT

COVID 19 has made a huge impact on the society, the new restriction has been imposed as in the number of users allowed in a particular room in offices, shops, etc. to maintain social distancing, along with social distancing regular temperature check at entrances of malls, the office is mandatory. In this project we stimulate a room where such necessary precautions are taken, we make use of a laser diode and receiver to detect the entrance of a person, when the project detect entrance it will check the temperature of the person if the temperature is less than the set temperature the person is allowed entry otherwise the entry is denied. only a pre-determined number of people are allowed in the room. The allowed temperature, the number of people allowed in the room as well as the number of people actively present in the room can be set/viewed using a Bluetooth App.

2. INTRODUCTION

Fever is one of the most important symptoms of COVID-19, but due to the contagious effect, its measurement can become a serious problem, so it is important to perform the temperature detection of patients very quickly and possibly without any contact. On the other hand, both epidemiological and laboratory studies have revealed that ambient temperature could affect the survival and spread of Coronavirus, so that a continuous monitoring of temperature, regarding both ambient and body temperature, is an essential task to be performed in the contrast of COVID-19. The adoption of thermographic systems in the framework of pandemic situations, such as COVID-19, can be decisive for initial temperature assessment devoted to medical purposes, namely:

- Initial human temperature screening during the triage process in a public health emergency, to determine the significance of fever and elevated temperature with respect to possible affection;
- Temperature assessment within high throughput areas, such as business structures, airports, etc.

The additional monitoring of ambient temperature can be strongly helpful to improve the accuracy in the detection of human body temperature, the measurement of which can be affected by the environmental humidity. Furthermore, recent studies have confirmed the effects of environmental temperature and humidity values in the increased risk of COVID-19 transmission

3. LITERATURE SURVEY

“Sensors for Body Temperature Measurement and Monitoring in a Time of Pandemic” Covid-19 has demonstrated how vulnerable our society remains, when it comes to managing a pandemic. There is an urgent need to develop and evaluate new technologies, materials, and methods for rapid measurement and continuous monitoring of body temperature, and accurate detection of elevated body temperature as a symptom of conditions that may lead to a pandemic. This Special Issue of the IEEE Sensors Journal is devoted to the presentation of innovations and improvements in the field of human and animal body temperature measurement and monitoring including theory, design, modelling, fabrication, configuration, characterization, manufacturing, data analysis, and applications. Original research contributions and review papers are sought in areas.

including:

- Body Temperature Sensor (BTS) Theory, Design, and Modelling
- BTS Materials, Processing, Fabrication, and Packaging
- BTS Types (NTC Thermistor, RTD, Semiconductor, MEMS, Infrared)
- New BTS Technologies
- Wearable and Implantable BTS and Systems
- Remote BTS and Systems
- BTS Characterization and Evaluation
- BTS Signal Processing and Data Collection
- Low Power and Miniaturized BTS
- BTS Networks, IoT, and Data Security
- Algorithms and Tools for Body Temperature Screening
- Elevated Body Temperature Detection Using Artificial Intelligence
- Health Applications of Body Temperature Measurement and Monitoring

4. BLOCK DIAGRAM

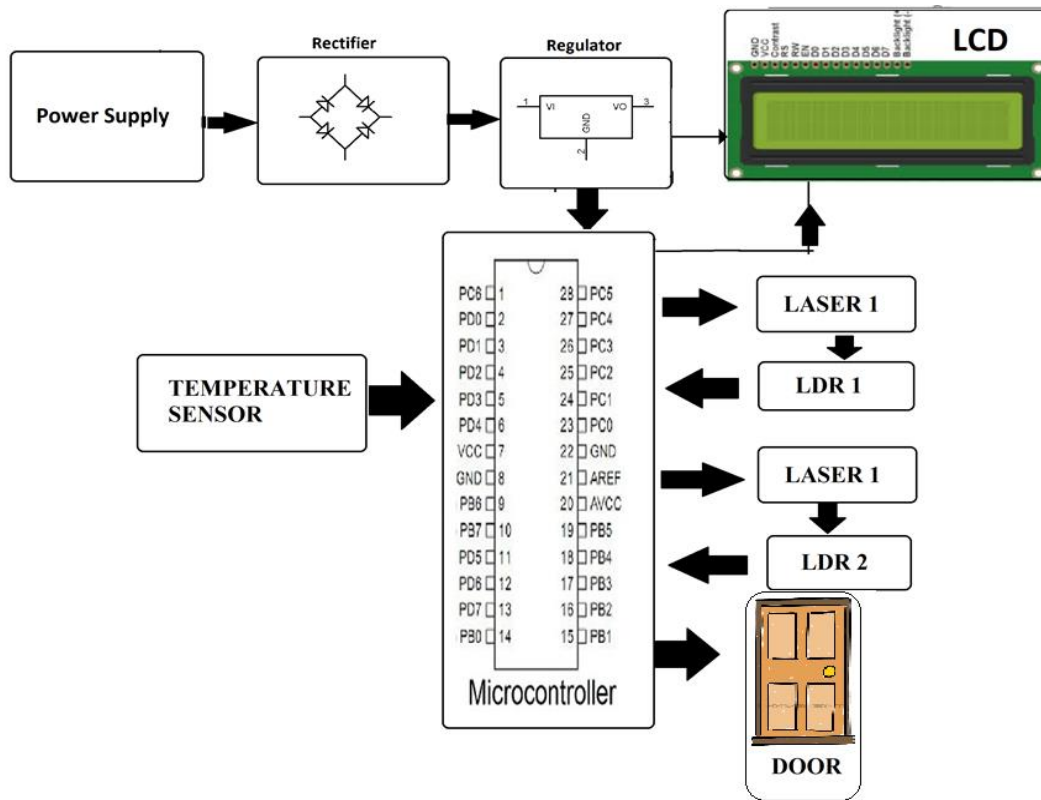


FIG (1) BLOCK DIAGRAM

4.1 Operation Of The System

Regulator and rectifier is part of the power supply unit of this system, it's powers up the LCD display as well as the microcontroller. The Non-Contact Infrared Temperature Sensor is interfaced with the ucontroller atmega 328p microcotroller to work accordingly. The LDRs and lasers are also interfaced with the ucontroller to execute as per the program. The temp sensed by the sensor gives out the result on the LCD screen and along with this execution the laser related to that result blinks and also the door works accordingly

5. COMPONENT LIST

- 1) Atmega 328p Microcontroller
- 2) Non Contact Temperature Sensor
- 3) Bluetooth Module HC-05
- 4) DC Motor
- 5) 16x2 LCD Display
- 6) Vtg Regulator IC
- 7) Resistors
- 8) Capacitors
- 9) Transistors
- 10) Cables and Connectors
- 11) Diodes
- 12) PCB and Breadboards

13) LED

14) Transformer/Adapter

15) Push Buttons

16) Buzzer

17) IC

18) IC Sockets

19) Rectifier

20) Filter

6. SPECIFICATION OF COMPOENTS

6.1 ATmega 328 p Microcontroller:

The Atmel ATmega328P is a 32K 8-bit microcontroller based on the AVR architecture. Many instructions are executed in a single clock cycle providing a throughput of almost 20 MIPS at 20MHz. The ATMEGA328-PU comes in an PDIP 28 pin package and is suitable for use on our 28 pin AVR Development Board.

The computer on one hand is designed to perform all the general purpose tasks on a single machine like you can use a computer to run a software to perform calculations or you can use a computer to store some multimedia file or to access internet through the browser, whereas the microcontrollers are meant to perform only the specific tasks, for e.g., switching the AC off automatically when room temperature drops to a certain defined limit and again turning it ON when temperature rises above the defined limit.

There are number of popular families of microcontrollers which are used in different applications as per their capability and feasibility to perform the desired task, most common of these are 8051, AVR and PIC microcontrollers. In this we will introduce you with AVR family of microcontrollers.



FIG (2) ATMEGA 328P MICROCONTROLLER

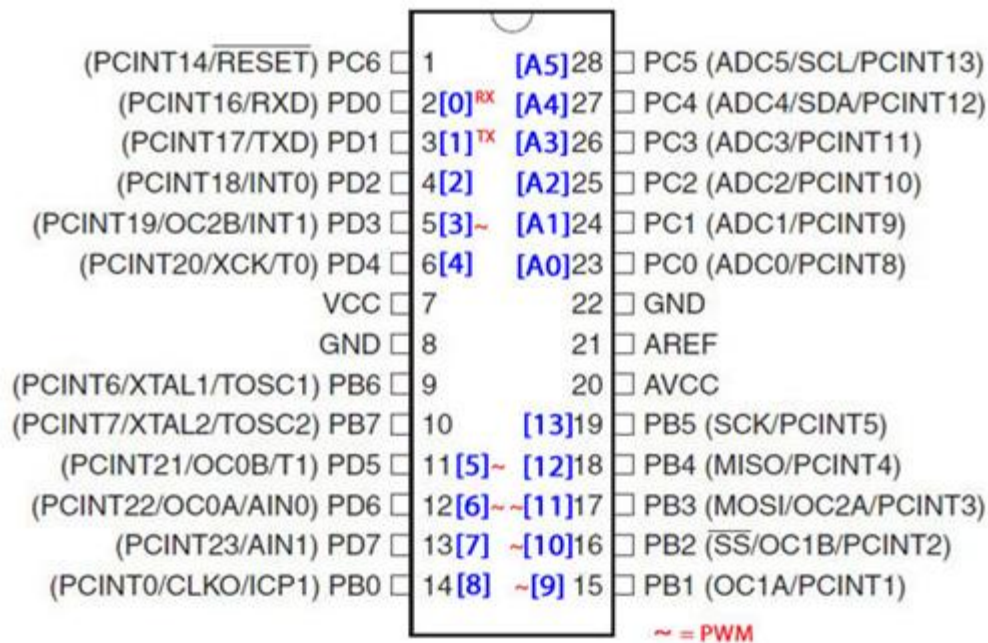
6.1.1 Features include:

- High Performance, Low Power Design
- 8-Bit Microcontroller Atmel® AVR® advanced RISC architecture
 - 131 Instructions most of which are executed in a single clock cycle
 - Up to 20 MIPS throughput at 20 MHz
 - 32 x 8 working registers
 - 2 cycle multiplier
- Memory Includes
 - 32KB of of programmable FLASH
 - 1KB of EEPROM
 - 2KB SRAM
 - 10,000 Write and Erase Cycles for Flash and 100,000 for EEPROM
 - Data retention for 20 years at 85°C and 100 years at 25°C
 - Optional boot loader with lock bits
 - In System Programming (ISP) by via boot loader
 - True Read-While-Write operation
 - Programming lock available for software security
- Features Include
 - 2 x 8-bit Timers/Counters each with independent prescaler and compare modes
 - A single 16-bit Timer/Counter with an independent prescaler, compare and capture modes
 - Real time counter with independent oscillator
 - 10 bit, 6 channel analog to digital Converter
 - 6 pulse width modulation channels
 - Internal temperature sensor
 - Serial USART (Programmable)
 - Master/Slave SPI Serial Interface - (Philips I2C compatible)
 - Programmable watchdog timer with independent internal oscillator
 - Internal analog comparator
 - Interrupt and wake up on pin change
- Additional Features Features
 - Internal calibrated oscillator
 - Power on reset and programmable brown out detection
 - External and internal interrupts
 - 6 sleep modes including idle, ADC noise reduction, power save, power down, standby, and extended standby

- I/O and Package
 - 23 programmable I/O lines
 - 28 pin PDIP package
- Operating voltage:
 - 1.8 - 5.5V
- Operating temperature range:
 - 40°C to 85°C
- Speed Grades:
 - 0-4 MHz at 1.8-5.5V
 - 0-10 MHz at 2.7-5.5V
 - 0-20 MHz at 4.5-5.5V

Flash:	32 KBytes
EEPROM:	1 KBytes
SRAM:	2 KBytes
Max I/O Pins:	23
Frequency Max:	20 MHz
VCC:	1.8-5.5
10-bit A/D Channels:	6
Analog Comparator:	Yes
16-bit Timers:	1
8-bit Timer:	2
Brown Out Detector:	Yes
Ext Interrupts:	2
Hardware Multiplier:	Yes
Interrupts:	26
ISP:	Yes
On Chip Oscillator:	Yes
PWM Channels:	6
RTC:	Yes
Self Program Memory:	Yes
SPI:	1
TWI:	Yes
UART:	1
Watchdog:	Yes

TABLE NO 1



In our days, there have been many advancement in the field of Electronics and many cutting edge technologies are being developed every day, but still 8 bit microcontrollers have its own role in the digital electronics market dominated by 16-32 & 64 bit digital devices. Although powerful microcontrollers with higher processing capabilities exist in the market, 8bit microcontrollers still hold its value because of their easy-to-understand-operation, very much high popularity, ability to simplify a digital circuit, low cost compared to features offered, addition of many new features in a single IC and interest of manufacturers and consumers.

Today's microcontrollers are much different from what it were in the initial stage, and the number of manufacturers are much more in count than it was a decade or two ago. At present some of the major manufacturers are Microchip (publication: PIC microcontrollers), Atmel (publication: AVR microcontrollers), Hitachi, Phillips, Maxim, NXP, Intel etc. Our interest is upon ATmega32. It belongs to Atmel's AVR series micro controller family. Let's see the features.

PIN count: Atmega32 has got 40 pins. Two for Power (pin no.10: +5v, pin no. 11: ground), two for oscillator (pin 12, 13), one for reset (pin 9), three for providing necessary power and reference voltage to its internal ADC, and 32 (4×8) I/O pins.

About I/O pins: ATmega32 is capable of handling analogue inputs. Port A can be used as either DIGITAL I/O Lines or each individual pin can be used as a

single input channel to the internal ADC of ATmega32, plus a pair of pins AREF, AVCC & GND (refer to ATmega32 datasheet) together can make an ADC channel.

No pins can perform and serve for two purposes (for an example: Port A pins cannot work as a Digital I/O pin while the Internal ADC is activated) at the same time. It's the programmers responsibility to resolve the conflict in the circuitry and the program. Programmers are advised to have a look to the priority tables and the internal configuration from the datasheet.

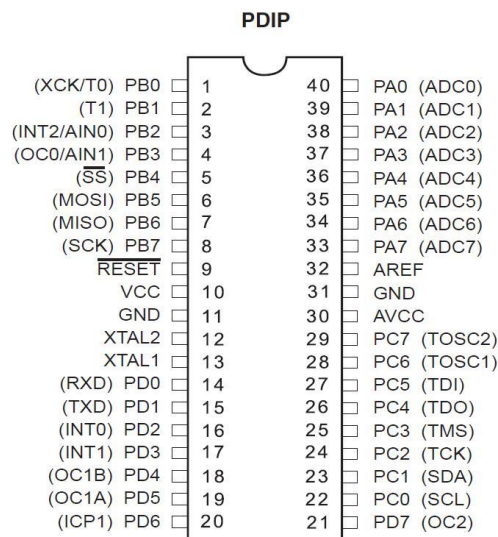
Digital I/O pins: ATmega32 has 32 pins (4portsx8pins) configurable as Digital I/O pins.

Timers: 3 Inbuilt timer/counters, two 8 bit (timer0, timer2) and one 16 bit (timer1).

ADC: It has one successive approximation type ADC in which total 8 single channels are selectable. They can also be used as 7 (for TQFP packages) or 2 (for DIP packages) differential channels. Reference is selectable, either an external reference can be used or the internal 2.56V reference can be brought into action. There external reference can be connected to the AREF pin.

Communication Options: ATmega32 has three data transfer modules embedded in it. They are

- Two Wire Interface
- USART
- Serial Peripheral Interface



Analog comparator: On-chip analog comparator is available. An interrupt is assigned for different comparison result obtained from the inputs.

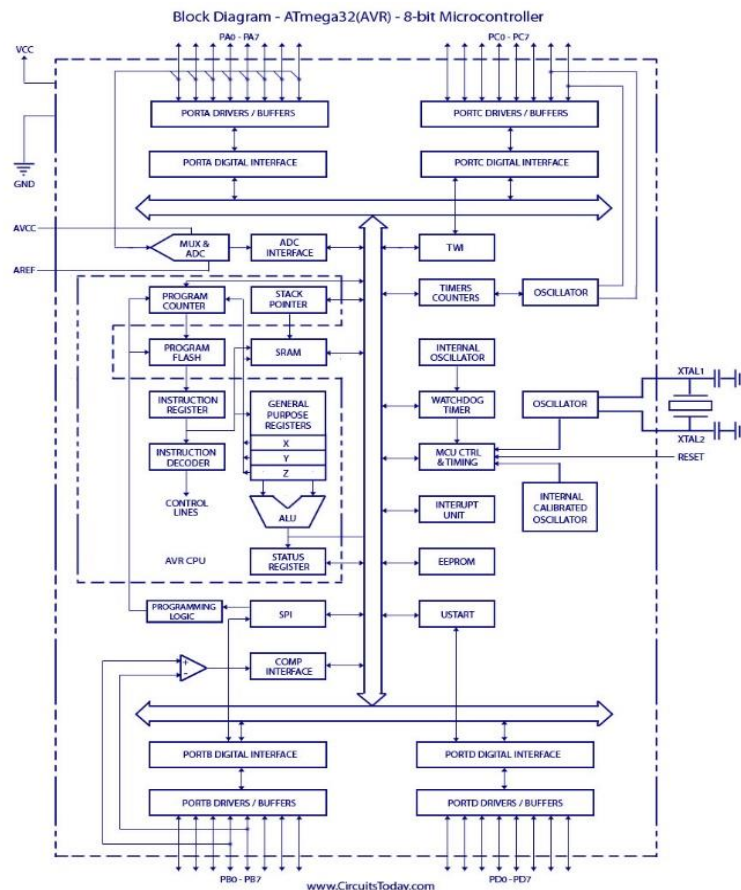
External Interrupt: 3 External interrupt is accepted. Interrupt sense is configurable.

Memory: It has 32Kbytes of In-System Self-programmable Flash program memory, 1024 Bytes EEPROM, 2Kbytes Internal SRAM. Write/Erase Cycles: 10,000 Flash / 100,000 EEPROM.

Clock: It can run at a frequency from 1 to 16 MHz. Frequency can be obtained from external Quartz Crystal, Ceramic crystal or an R-C network. Internal calibrated RC oscillator can also be used.

Debug: JTAG boundary scan facilitates on chip debug.

Programming: Atmega32 can be programmed either by In-System Programming via Serial peripheral interface or by Parallel programming. Programming via JTAG interface is also possible. Programmer must ensure that SPI programming and JTAG are not be disabled using fuse bits; if the programming is supposed to be done using SPI or JTAG.



6.2 Non-Contact Infrared Temperature Sensor Module:

The an infrared thermometer for non-contact temperature measurements. Both the IR sensitive thermopile detector chip and the signal conditioning ASIC are integrated in the same TO-39 can. Integrated into the MLX90614 are a low noise amplifier, 17-bit ADC and powerful DSP unit thus achieving high accuracy and resolution of the thermometer.

The thermometer comes factory calibrated with a digital SMBus output giving full access to the measured temperature in the complete temperature range(s) with a resolution of 0.02°C . The user can configure the digital output to be pulse width modulation (PWM).

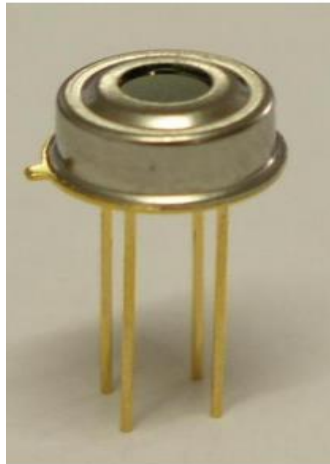
As a standard, the 10-bit PWM is configured to continuously transmit the measured temperature in range of -20 to 120°C , with an output resolution of 0.14°C .



FIG (3) INFRARED TEMPERATURE SENSOR MODULE

The operation of the MLX90614 is controlled by an internal state machine, which controls the measurements and calculations of the object and ambient temperatures and does the post-processing of the temperatures to output them through the PWM output or the SMBus compatible interface. The ASSP supports 2 IR sensors (second one not implemented in the MLX90614xAx). The output of the IR sensors is amplified by a low noise low offset chopper amplifier with

programmable gain, converted by a Sigma Delta modulator to a single bit stream and fed to a powerful DSP for further processing. The signal is treated by programmable (by means of EEPROM content) FIR and IIR low pass filters for further reduction of the band width of the input signal to achieve the desired noise performance and refresh rate. The output of the IIR filter is the measurement result and is available in the internal RAM. 3 different cells are available: One for the on-board temperature sensor (on chip PTAT or PTC) and 2 for the IR sensors. Based on results of the above measurements, the corresponding ambient temperature T_a and object temperatures T_o are calculated. Both calculated temperatures have a resolution of $0.01\text{ }^{\circ}\text{C}$. The data for T_a and T_o can be read in two ways: Reading RAM cells dedicated for this purpose via the 2-wire interface (0.02°C resolution, fixed ranges), or through the PWM digital output (10 bit resolution, configurable range). In the last step of the measurement cycle, the measured T_a and T_o are rescaled to the desired output resolution of the PWM) and the recalculated data is loaded in the registers of the PWM state machine, which creates a constant frequency with a duty cycle representing the measured data.



6.2.1 Features and Benefits

- Small size, low cost
- Easy to integrate
- Factory calibrated in wide temperature range: -40 to $125\text{ }^{\circ}\text{C}$ for sensor temperature and -70 to $380\text{ }^{\circ}\text{C}$ for object temperature.
- High accuracy of 0.5°C over wide temperature range ($0..+50^{\circ}\text{C}$ for both T_a and T_o)
- High (medical) accuracy calibration optional

- Measurement resolution of 0.02°C
- Single and dual zone versions
- SMBus compatible digital interface
- Customizable PWM output for continuous reading
- Available in 3V and 5V versions
- Simple adaptation for 8 to 16V applications
- Power saving mode
- Different package options for applications and measurements versatility
- Automotive grade

6.2.2 Applications Examples:

- High precision non-contact temperature measurements
- Thermal Comfort sensor for Mobile Air Conditioning control system
- Temperature sensing element for residential, commercial and industrial building air conditioning
- Windshield defogging
- Automotive blind angle detection
- Industrial temperature control of moving parts
- Temperature control in printers and copiers
- Home appliances with temperature control
- Healthcare
- Livestock monitoring
- Movement detection
- Multiple zone temperature control – up to 100 sensors can be read via common 2 wires
- Thermal relay/alert
- Body temperature measurement

6.3 Bluetooth Module HC-05:

HC-05 module is an easy to use Bluetooth SPP (Serial Port Protocol) module, designed for transparent wireless serial connection setup. Serial port Bluetooth module is fully qualified Bluetooth V2.0+EDR (Enhanced Data Rate) 3Mbps Modulation with complete 2.4GHz radio transceiver and baseband. It uses CSR Bluecore 04-External single chip Bluetooth system with CMOS technology and with AFH(Adaptive Frequency Hopping Feature). It has the footprint as small as 12.7mmx27mm. Hope it will simplify your overall design/development cycle.

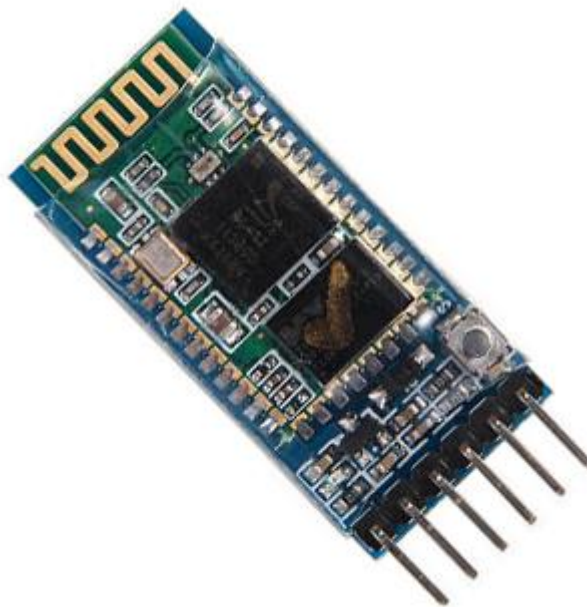


FIG (4) BLUETOOTH MODULE HC-05

6.3.1 Hardware features:

- Typical -80dBm sensitivity
- Up to +4dBm RF transmit power
- Low Power 1.8V Operation ,1.8 to 3.6V I/O
- PIO control
- UART interface with programmable baud rate
- With integrated antenna
- With edge connector

6.3.2 Software features:

- Default Baud rate: 38400, Data bits:8, Stop bit:1,Parity:No parity, Data control: has. Supported baud rate: 9600,19200,38400,57600,115200,230400,460800.
- Given a rising pulse in PIO0, device will be disconnected.
- Status instruction port PIO1: low-disconnected, high-connected
- PIO10 and PIO11 can be connected to red and blue led separately. When master and slave are paired, red and blue led blinks 1time/2s in interval, while disconnected only blue led blinks 2times/s.
- Auto-connect to the last device on power as default.
- Permit pairing device to connect as default.
- Auto-pairing PINCODE:”0000” as default
- Auto-reconnect in 30 min when disconnected as a result of beyond the range of connection.

6.3.3 Pin Definition:

Pin	Description	Function
VCC	+5V	Connect to +5V
GND	Ground	Connect to Ground
TXD	UART_TXD, Bluetooth serial signal sending PIN	Connect with the MCU's (Microcontroller and etc) RXD PIN
RXD	UART_RXD, Bluetooth serial signal receiving PIN	Connect with the MCU's (Microcontroller and etc) TXD PIN
KEY	Mode switch input	If it is input low level or connect to the air, the module is at paired or communication mode. If it's input high level, the module will enter to AT mode.

TABLE NO 2

6.4 DC Motor:

DC Motor – 100RPM – 12Volts geared motors are generally a simple DC motor with a gearbox attached to it. This can be used in all-terrain robots and variety of robotic applications. These motors have a 3 mm threaded drill hole in the middle of the shaft thus making it simple to connect it to the wheels or any other mechanical assembly.

100 RPM 12V DC geared motors widely use for robotics applications. Very easy to use and available in standard size. Also, you don't have to spend a lot of money to control motors with an Arduino or compatible board. The most popular L298N H-bridge module with onboard voltage regulator motor driver can be used with this motor that has a voltage of between 5 and 35V DC or you can choose the most precise motor driver module from the wide range available in our Motor drivers category as per your specific requirements.

Nut and threads on the shaft to easily connect and internally threaded shaft for easily connecting it to the wheel. DC Geared motors with robust metal gearbox for heavy-duty applications, available in the wide RPM range and ideally suited for robotics and industrial applications. Very easy to use and available in standard size. Nut and threads on the shaft to easily connect and internally threaded shaft for easily connecting it to the wheel.

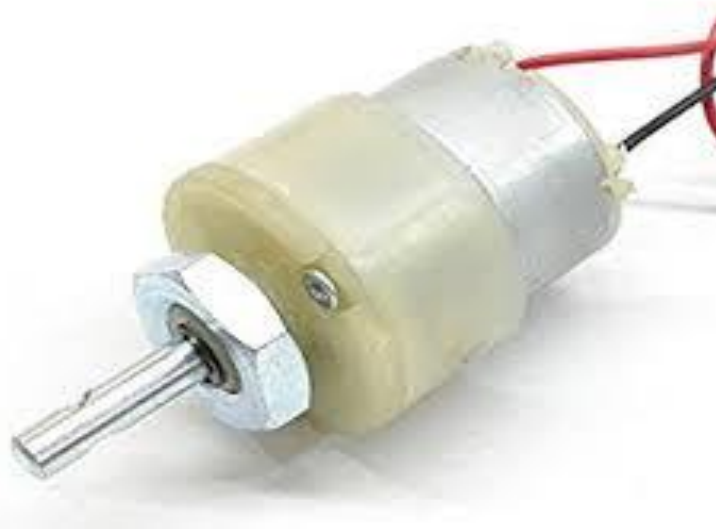


FIG (5) DC MOTOR

6.4.1 Specifications:

- DC supply: 4 to 12V
- RPM: 100 at 12V
- Total length: 46mm
- Motor diameter: 36mm
- Motor length: 25mm
- Brush type: Precious metal
- Gear head diameter: 37mm
- Gear head length: 21mm
- Output shaft: Centred
- Shaft diameter: 6mm
- Shaft length: 22mm
- Gear assembly: Spur
- Motor weight: 100gms

6.4.2 Motor performance in terms of RPM and no load current as a function of input voltage

Voltage (V)	RPM (No Load)	Current (A)
4	30	0.016
5	36	0.017
6	46	0.019
7	53	0.020
8	60	0.022
9	68	0.024
10	76	0.027
11	83	0.030
12	92	0.035

TABLE NO 3

6.5 16x2 LCD Display :

LCD modules are very commonly used in most embedded projects, the reason being its cheap price, availability and programmer friendly. Most of us would have come across these displays in our day to day life, either at PCO's or calculators. The appearance and the pinouts have already been visualized above now let us get a bit technical.

16×2 LCD is named so because; it has 16 Columns and 2 Rows. There are a lot of combinations available like, 8×1, 8×2, 10×2, 16×1, etc. but the most used one is the 16×2 LCD. So, it will have (16×2=32) 32 characters in total and each character will be made of 5×8 Pixel Dots.

we know that each character has (5×8=40) 40 Pixels and for 32 Characters we will have (32×40) 1280 Pixels. Further, the LCD should also be instructed about the Position of the Pixels. Hence it will be a hectic task to handle everything with the help of MCU, hence an Interface IC like HD44780 is used, which is mounted on the backside of the LCD Module itself. The function of this IC is to get the Commands and Data from the MCU and process them to display meaningful information onto our LCD Screen.

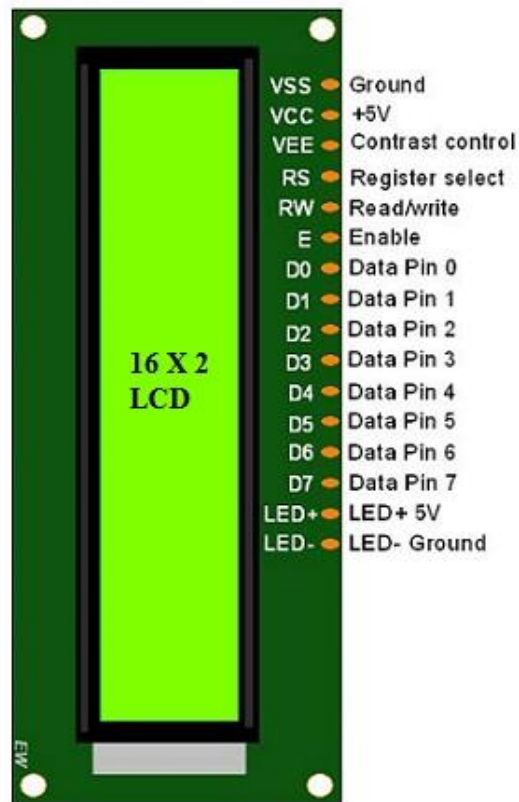


FIG (6) 16*2 LCD DISPLAY

6.5.1 PINOUTS:

Pin1 (Ground/Source Pin): This is a GND pin of display, used to connect the GND terminal of the microcontroller unit or power source. Pin2 (VCC/Source Pin): This is the voltage supply pin of the display, used to connect the supply pin of the power source. Pin3 (V0/VEE/Control Pin): This pin regulates the

difference of the display, used to connect a changeable POT that can supply 0 to 5V. Pin4 (Register Select/Control Pin): This pin toggles among command or data register, used to connect a microcontroller unit pin and obtains either 0 or 1(0 = data mode, and 1 = command mode).Pin5 (Read/Write/Control Pin): This pin toggles the display among the read or writes operation, and it is connected to a microcontroller unit pin to get either 0 or 1 (0 = Write Operation, and 1 = Read Operation). Pin 6 (Enable/Control Pin): This pin should be held high to execute Read/Write process, and it is connected to the microcontroller unit & constantly held high.Pins 7-14 (Data Pins): These pins are used to send data to the display. These pins are connected in two-wire modes like 4-wire mode and 8-wire mode. In 4-wire mode, only four pins are connected to the microcontroller unit like 0 to 3, whereas in 8-wire mode, 8-pins are connected to microcontroller unit like 0 to 7. Pin15 (+ve pin of the LED): This pin is connected to +5V Pin 16 (-ve pin of the LED): This pin is connected to GND.



6.6 Voltage Regulator 7805:

The LM78XX/LM78XXA series of three-terminal positive regulators are available in the TO-220/D-PAK package and with several fixed output voltages, making them useful in a Wide range of applications. Each type employs internal current limiting, thermal shutdown and safe operating area protection, making it essentially indestructible. If adequate heat sinking is provided, they can deliver over 1A output Current. Although designed primarily as fixed voltage regulators, these devices can be used with external components to obtain adjustable voltages and currents.

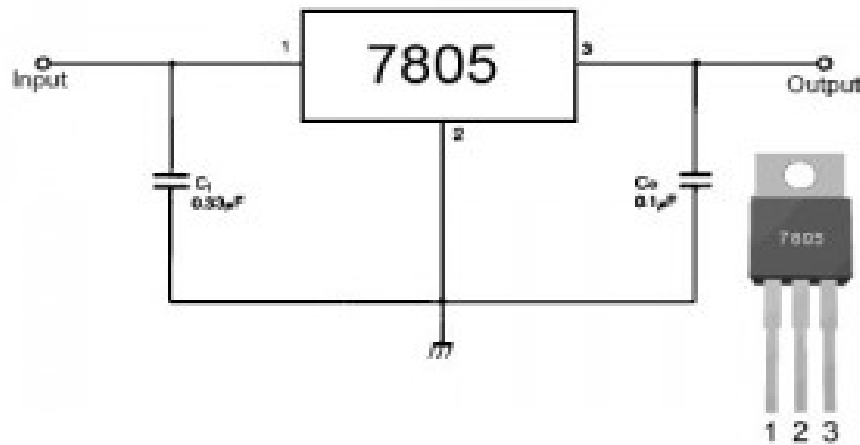


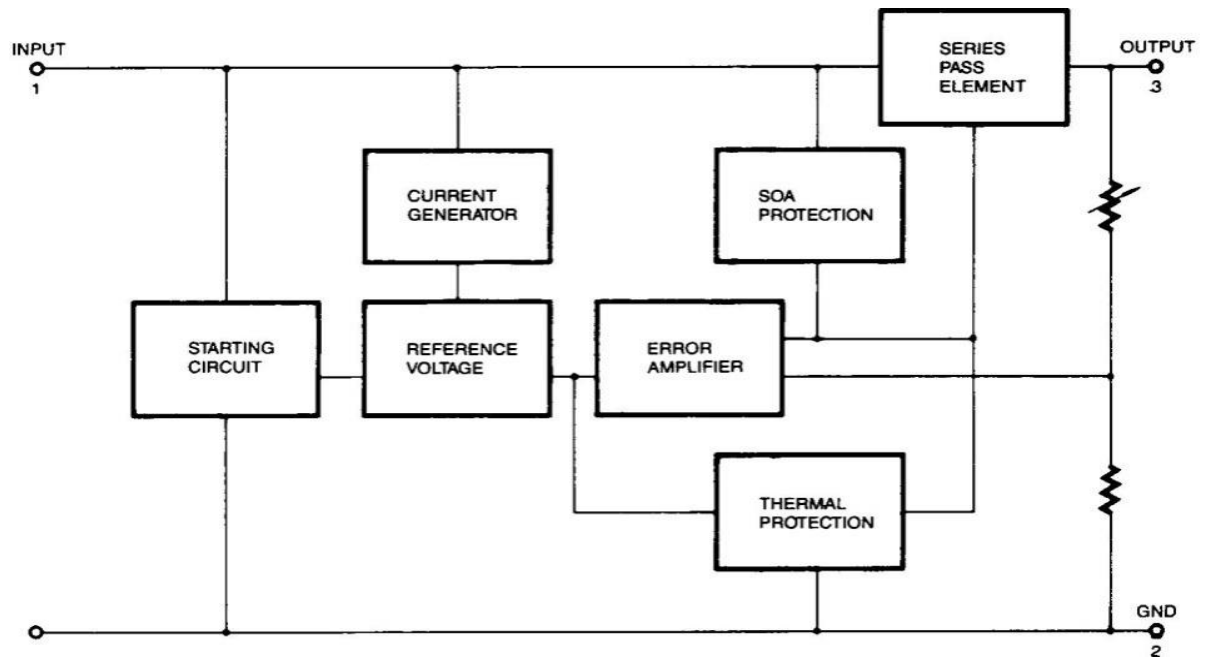
FIG (7) VTG REGULATOR 7805

6.6.1 Absolute Maximum Ratings:

Parameter	Symbol	Value	Unit
Input Voltage (for $V_O = 5V$ to $18V$)	V_I	35	V
(for $V_O = 24V$)	V_I	40	V
Thermal Resistance Junction-Cases (TO-220)	$R_{\theta JC}$	5	$^{\circ}C/W$
Thermal Resistance Junction-Air (TO-220)	$R_{\theta JA}$	65	$^{\circ}C/W$
Operating Temperature Range (KA78XX/A/R)	T_{OPR}	$0 \sim +125$	$^{\circ}C$
Storage Temperature Range	T_{STG}	$-65 \sim +150$	$^{\circ}C$

TABLE NO 4

6.6.2 Internal Block Diagram:



6.6.3 Features:

- Output Current up to 1A.
- Output Voltages of 5, 6, 8, 9, 10, 12, 15, 18, 24V.
- Thermal Overload Protection.
- Short Circuit Protection.
- Output Transistor Safe Operating Area Protection.

6.7 RESISTOR:

A resistor is a two-terminal electronic component designed to oppose an electric current by producing a voltage drop between its terminals in proportion to the current, that is, in accordance with Ohm's law:

$$V = IR$$

Resistors are used as part of electrical networks and electronic circuits. They are extremely commonplace in most electronic equipment. Practical resistors can be made of various compounds and films, as well as resistance wire (wire made of a high-resistivity alloy, such as nickel/chrome).



FIG (8) RESISTORS

The primary characteristics of resistors are their resistance and the power they can dissipate. Other characteristics include temperature coefficient, noise, and inductance. Less well-known is critical resistance, the value below which power dissipation limits the maximum permitted current flow, and above which the limit is applied voltage. Critical resistance depends upon the materials constituting the resistor as well as its physical dimensions; it's determined by design.

Resistors can be integrated into hybrid and printed circuits, as well as integrated circuits. Size, and position of leads (or terminals) are relevant to equipment designers; resistors must be physically large enough not to overheat when dissipating their power.

A resistor is a two-terminal passive electronic component which implements electrical resistance as a circuit element. When a voltage V is applied across the

terminals of a resistor, a current I will flow through the resistor in direct proportion to that voltage. The reciprocal of the constant of proportionality is known as the resistance R , since, with a given voltage V , a larger value of R further "resists" the flow of current I as given by Ohm's law:

$$I = \frac{V}{R}$$

Resistors are common elements of electrical networks and electronic circuits and are ubiquitous in most electronic equipment. Practical resistors can be made of various compounds and films, as well as resistance wire (wire made of a high-resistivity alloy, such as nickel-chrome). Resistors are also implemented within integrated circuits, particularly analog devices, and can also be integrated into hybrid and printed circuits.

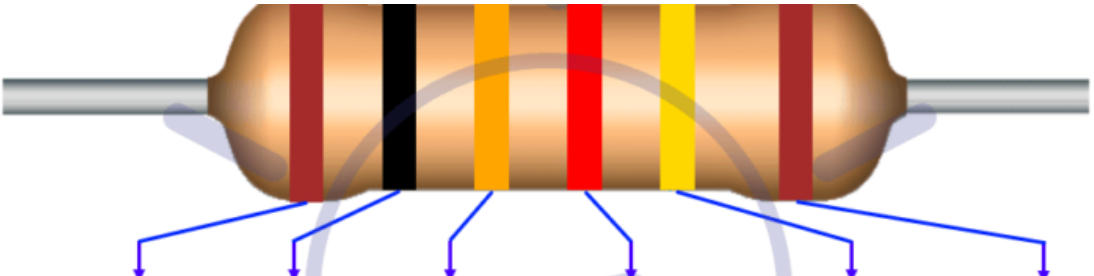
The electrical functionality of a resistor is specified by its resistance: common commercial resistors are manufactured over a range of more than 9 orders of magnitude. When specifying that resistance in an electronic design, the required precision of the resistance may require attention to the manufacturing tolerance of the chosen resistor, according to its specific application. The temperature coefficient of the resistance may also be of concern in some precision applications. Practical resistors are also specified as having a maximum power rating which must exceed the anticipated power dissipation of that resistor in a particular circuit: this is mainly of concern in power electronics applications. Resistors with higher power ratings are physically larger and may require heat sinking. In a high voltage circuit, attention must sometimes be paid to the rated maximum working voltage of the resistor.

The series inductance of a practical resistor causes its behaviour to depart from ohms law; this specification can be important in some high-frequency applications for smaller values of resistance. In a low-noise amplifier or pre-amp the noise characteristics of a resistor may be an issue. The unwanted inductance, excess noise, and temperature coefficient are mainly dependent on the technology used in manufacturing the resistor. They are not normally specified individually for a particular family of resistors manufactured using a particular technology. A family of discrete resistors is also characterized according to its form factor, that is, the size of the device and position of its leads (or terminals) which is relevant in the practical manufacturing of circuits using them.

6.7.1 FEATURES:

- RoHS compliant - Products with lead free terminations meet RoHS requirements - Pb-glass contained in electrodes, resistor element and glass are exempted by RoHS
- Reducing environmentally hazardous wastes
- High component and equipment reliability
- Saving of PCB space
- None forbidden-materials used in products/production
- Halogen Free Epoxy

6.7.2 Resistor Color Code Calculator:



COLOR	1 st DIGIT	2 nd DIGIT	3 rd DIGIT	MULTIPLIER	TOLERANCE	TEMP.Co
BLACK	0	0	0	1Ω		
BROWN	1	1	1	10Ω	± 1% (F)	100
RED	2	2	2	100Ω	± 2% (G)	50
ORANGE	3	3	3	1kΩ	± 3%	15
YELLOW	4	4	4	10kΩ	± 4%	25
GREEN	5	5	5	100kΩ	± 0.5% (D)	
BLUE	6	6	6	1MΩ	± 0.25% (C)	10
VIOLET	7	7	7	10MΩ	± 0.10% (B)	5

6.8 CAPACITORS:

A capacitor or condenser is a passive electronic component consisting of a pair of conductors separated by a dielectric. When a voltage potential difference exists between the conductors, an electric field is present in the dielectric. This field stores energy and produces a mechanical force between the plates. The effect is greatest between wide, flat, parallel, narrowly separated conductors.

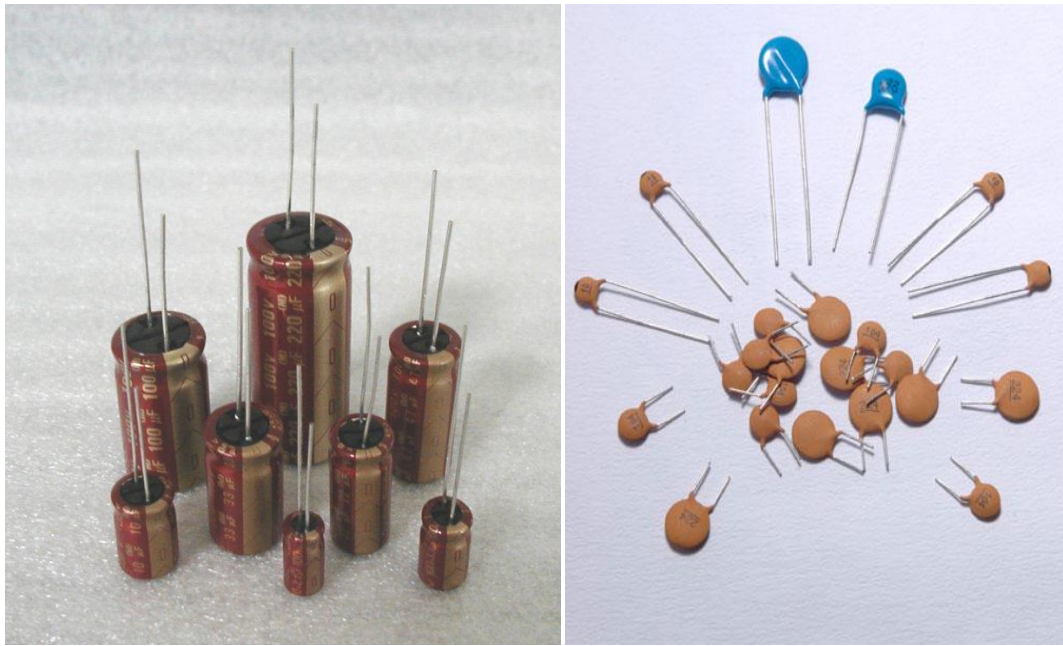


FIG (9) CAPACITORS

An ideal capacitor is characterized by a single constant value, capacitance, which is measured in farads. This is the ratio of the electric charge on each conductor to the potential difference between them. In practice, the dielectric between the plates passes a small amount of leakage current. The conductors and leads introduce an equivalent series resistance and the dielectric has an electric field strength limit resulting in a breakdown voltage.

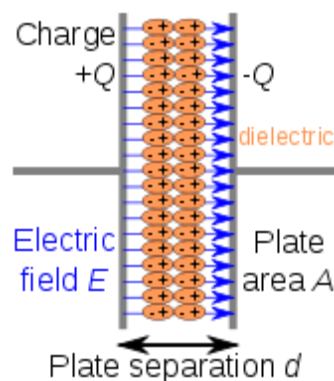
The properties of capacitors in a circuit may determine the resonant frequency and quality factor of a resonant circuit, power dissipation and operating frequency in a digital logic circuit, energy capacity in a high-power system, and many other important aspects.

A capacitor (formerly known as condenser) is a device for storing electric charge. The forms of practical capacitors vary widely, but all contain at least two conductors separated by a non-conductor. Capacitors used as parts of electrical systems, for example, consist of metal foils separated by a layer of insulating film.

Capacitors are widely used in electronic circuits for blocking direct current while allowing alternating current to pass, in filter networks, for smoothing the output of power supplies, in the resonant circuits that tune radios to particular frequencies and for many other purposes.

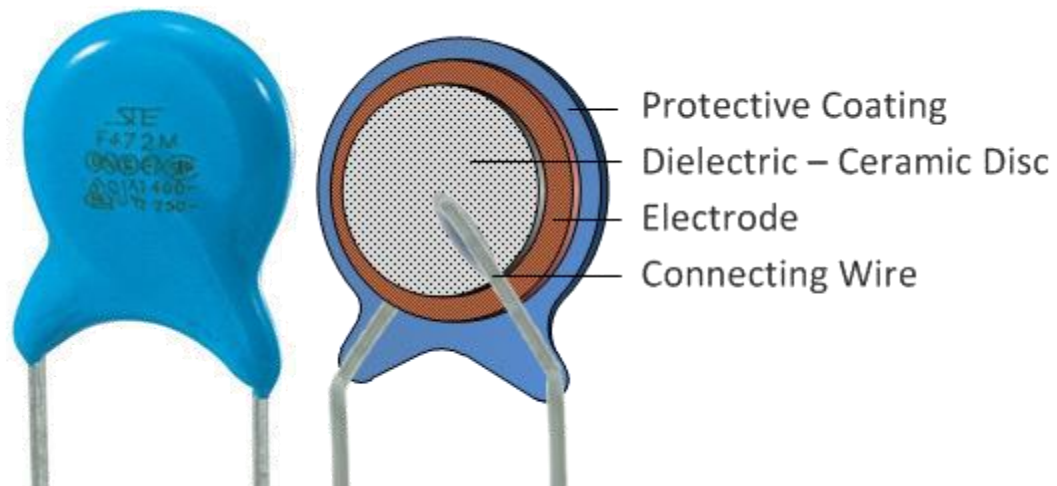
A capacitor is a passive electronic component consisting of a pair of conductors separated by a dielectric (insulator). When there is a potential difference (voltage) across the conductors, a static electric field develops in the dielectric that stores energy and produces a mechanical force between the conductors. An ideal capacitor is characterized by a single constant value, capacitance, measured in farads. This is the ratio of the electric charge on each conductor to the potential difference between them.

The capacitance is greatest when there is a narrow separation between large areas of conductor, hence capacitor conductors are often called "plates", referring to an early means of construction. In practice the dielectric between the plates passes a small amount of leakage current and also has an electric field strength limit, resulting in a breakdown voltage, while the conductors and leads introduce an undesired inductance and resistance.



6.8.1 FEATURES:

- Ceramic singlelayer DC disc / AC disc capacitor
- High reliability
- High capacitance values up to 2 nF
- Small sizes
- Low losses
- Radial leads
- Material categorization: for definitions of compliance



6.8.2 APPLICATIONS:

- High voltage power supplies for x-ray sources and pulsed lasers
- Baggage scanner
- Medical x-ray
- Industrial laser

6.9 TRANSISTORS:

A transistor is a semiconductor device used to amplify or switch electronic signals and electrical power. Transistors are one of the basic building blocks of modern electronics. It is composed of semiconductor material usually with at least three terminals for connection to an external circuit.

A voltage or current applied to one pair of the transistor's terminals controls the current through another pair of terminals. Because the controlled (output) power can be higher than the controlling (input) power, a transistor can amplify a signal. Today, some transistors are packaged individually, but many more are found embedded in integrated circuits.

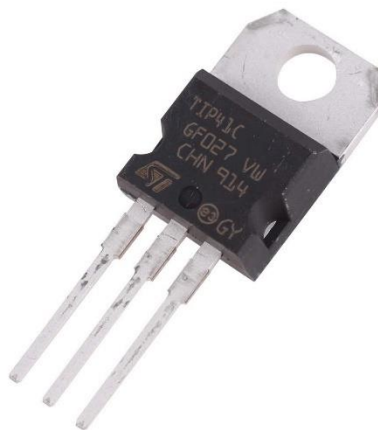


FIG (10) TRANSISTORS

Austro-Hungarian physicist Julius Edgar Lilienfeld proposed the concept of a field-effect transistor in 1926, but it was not possible to actually construct a working device at that time. The first working device to be built was a point-contact transistor invented in 1947 by American physicists John Bardeen and Walter Brattain while working under William Shockley at Bell Labs. The three shared the 1956 Nobel Prize in Physics for their achievement. The most widely used type of transistor is the metal–oxide–semiconductor field-effect transistor (MOSFET), which was invented

by Mohamed Atalla and Dawon Kahng at Bell Labs in 1959. Transistors revolutionized the field of electronics, and paved the way for smaller and cheaper radios, calculators, and computers, among other things.

Most transistors are made from very pure silicon, and some from germanium, but certain other semiconductor materials are sometimes used. A transistor may have only one kind of charge carrier, in a field-effect transistor, or may have two kinds of charge carriers in bipolar junction transistor devices. Compared with the vacuum tube, transistors are generally smaller and require less power to operate. Certain vacuum tubes have advantages over transistors at very high operating frequencies or high operating voltages. Many types of transistors are made to standardized specifications by multiple manufacturers.

6.9.1 Features:

- NPN Power Transistor.
- Low Frequency High Power amplifier.
- Continuous Collector current (I_C) is 3A.
- Collector-Emitter voltage (V_{CE}) is 50 V.
- Collector-Base voltage (V_{CB}) is 50.
- Emitter Base Voltage (V_{BE}) is 4V.
- DC Current Gain (h_{fe}): 35 to 320.
- Available in To-220 Package.

6.9.2 Applications:

- Class B amplifier designs
- Low Frequency Audio amplifier
- High Power audio amplifier
- Large signal amplification
- General purpose switching.

6.10 CABLES AND CONNECTORS:

6.10.1 Cables:

An electrical cable is an assembly of one or more **wires** running side by side or bundled, which is used to carry **electric current**.

A cable assembly is the composition of one or more electrical cables and their corresponding **connectors**.^[1] A cable assembly is not necessarily suitable for connecting two devices but can be a partial product (e.g. to be soldered onto a **printed circuit board** with a connector mounted to the housing). Cable assemblies can also take the form of a **cable tree** or **cable harness**, used to connect many terminals together.

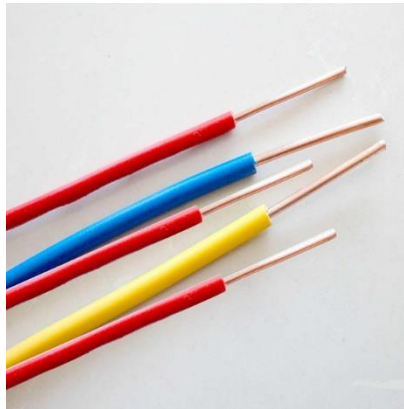


FIG (11) CABLES

6.10.2 Types Of Cable:

- Coaxial cable – used for radio frequency signals, for example in cable television distribution systems.
- Direct-buried cable
- Flexible cables
- Filled cable
- Heliac cable
- Non-metallic sheathed cable (or nonmetallic building wire, NM, NM-B)[2]
- Metallic sheathed cable (or armored cable, AC, or BX)[2]
- Multicore cable (consist of more than one wire and is covered by cable jacket)

6.10.3 Connectors:

Connectors are used to join subsections of circuits together. Usually, a connector is used where it may be desirable to disconnect the subsections at some future time: power inputs, peripheral connections, or boards which may need to be replaced.



FIG (12) 2 PIN RELIMATE

6.10.4 Type Of Connector:

- Rectangular Connectors.
- Circular Connectors.
- Terminal Blocks.
- Coaxial Connectors (RF)
- D-Sub, D-Shaped Connectors.
- Barrel Audio/Power Connectors.
- Pluggable Connectors.
- USB, DVI, HDMI Connectors.
- Terminal Connectors
- Backplane Connectors
- FFC,FPC(Flat Flex) Connectors
- Card Edge Connectors
- Sockets For ICs, Transistors

6.11 DIODE:

Diodes are used to convert AC into DC these are used as half wave rectifier or full wave rectifier. Three points must be kept in mind while using any type of diode.

- Maximum forward current capacity
- Maximum reverse voltage capacity
- Maximum forward voltage capacity



FIG (13) IN4001 DIODE

The number and voltage capacity of some of the important diodes available in the market are as follows:

Diodes of number IN4001, IN4002, IN4003, IN4004, IN4005, IN4006 and IN4007 have maximum reverse bias voltage capacity of 50V and maximum forward current capacity of 1 Amp.

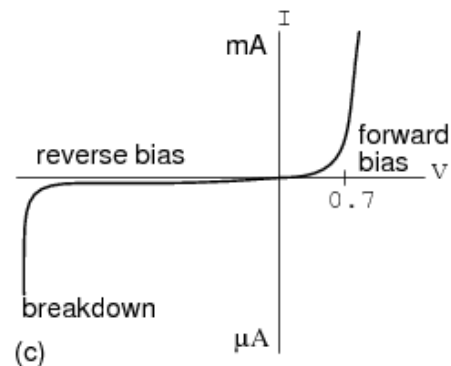
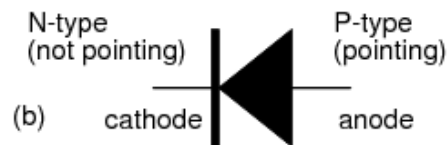
Diode of same capacities can be used in place of one another. Besides this diode of more capacity can be used in place of diode of low capacity but diode of low capacity cannot be used in place of diode of high capacity. For example, in place of IN4002; IN4001 or IN4007 can be used but IN4001 or IN4002 cannot be used in place of IN4007. The diode BY125 made by company BEL is equivalent of diode from IN4001 to IN4003. BY 126 is equivalent to diodes IN4004 to 4006 and BY 127 is equivalent to diode IN4007.

6.11.1 Features:

- Average forward current is 1A
- Non-repetitive Peak current is 30A
- Reverse current is 5 μ A.
- Peak repetitive Reverse voltage is 1000V
- Power dissipation 3W
- Available in DO-41 Package

6.11.2 Applications of Diode:

- Can be used to prevent reverse polarity problem
- Half Wave and Full Wave rectifiers
- Used as a protection device
- Current flow regulators



6.12 PCB AND BREADBOARD:

6.12.1 Printed Circuit Board(PCB):

A printed circuit board (PCB) mechanically supports and electrically connects electronic components using conductive tracks, pads and other features etched from one or more sheet layers of copper laminated onto and/or between sheet layers of a non-conductive substrate. Components are generally soldered onto the PCB to both electrically connect and mechanically fasten them to it. Printed circuit boards are used in all but the simplest electronic products. They are also used in some electrical products, such as passive switch boxes.



FIG (14) PRINTED CIRCUIT BOARD(PCB)

Alternatives to PCBs include wire wrap and point-to-point construction, both once popular but now rarely used. PCBs require additional design effort to lay out the circuit, but manufacturing and assembly can be automated. Electronic computer-aided design software is available to do much of the work of layout. Mass-producing circuits with PCBs is cheaper and faster than with other wiring methods, as components are mounted and wired in one operation. Large numbers of PCBs can be fabricated at the same time, and the layout only has to be done once. PCBs can also be made manually in small quantities, with reduced benefits.

PCBs can be single-sided (one copper layer), double-sided (two copper layers on both sides of one substrate layer), or multi-layer (outer and inner layers of copper, alternating with layers of substrate). Multi-layer PCBs allow for much higher component density, because circuit traces on the inner layers would otherwise take up surface space between components. The rise in popularity of

multilayer PCBs with more than two, and especially with more than four, copper planes was concurrent with the adoption of surface mount technology. However, multilayer PCBs make repair, analysis, and field modification of circuits much more difficult and usually impractical.

6.12.2 FEATURES:

- High quality of base material. ...
- Copper clad made tolerant as per international Standards. ...
- Hole wall thickness. ...
- **PCB's** with no track cut. ...
- Impeccably clean **PCB**. ...
- Treatments done on **PCB**. ...
- Positional Tolerance on **PCB**. ...
- Thickness of solder resist layer.

6.12.3 APPLICATION:

- **Monitors:** Personal and healthcare monitors including blood glucose monitors, heart rate and blood pressure monitors, and more.
- **Scanning Technology:** CT scanners and ultrasonic technology often use PCB-based electronics.
- **Control Systems:** The equipment controlling fluid infusion, flow rate and dispensation is electronically controlled.
- **Internal Devices:** Heart pacemakers and similar internal medical devices maintain patient health, run by minuscule PCBs at their centers.
- **Scientific Instrumentation:** Medical research uses numerous scientific instruments to study disease and test patient outcomes. These include electronic microscopes, photometers and control systems for generators and compressors, among other things.

6.12.4 Bread Board:

A breadboard, or protoboard, is a construction base for prototyping of electronics. Originally the word referred to a literal bread board, a polished piece of wood used when slicing bread.^[1] In the 1970s the solderless breadboard (a.k.a. plugboard, a terminal array board) became available and nowadays the term "breadboard" is commonly used to refer to these. Because the solderless breadboard does not require soldering, it is reusable. This makes it easy to use for creating temporary prototypes and experimenting with circuit design. For this reason, solderless breadboards are also popular with students and in technological education. Older breadboard types did not have this property. A stripboard (Veroboard) and similar prototyping printed circuit boards, which are used to build semi-permanent soldered prototypes or one-offs, cannot easily be reused. A variety of electronic systems may be prototyped by using breadboards, from small analog and digital circuits to complete central processing units (CPUs).



FIG (15) BREAD BOARD

Compared to more permanent circuit connection methods, modern breadboards have high parasitic capacitance, relatively high resistance, and less reliable connections, which are subject to jostle and physical degradation. Signaling is limited to about 10 MHz, and not everything works properly even well below that frequency. A common use in the system on a chip (SoC) era is to obtain an microcontroller (MCU) on a pre-assembled printed circuit board (PCB) which exposes an array of input/output (IO) pins in a header suitable to plug into a

breadboard, and then to prototype a circuit which exploits one or more of the MCU's peripherals, such as general-purpose input/output (GPIO), UART/USART serial transceivers, analog-to-digital converter (ADC), digital-to-analog converter (DAC), pulse-width modulation (PWM; used in motor control), Serial Peripheral Interface (SPI), or I²C.

Firmware is then developed for the MCU to test, debug, and interact with the circuit prototype. High frequency operation is then largely confined to the SoC's PCB. In the case of high speed interconnects such as SPI and I²C, these can be debugged at a lower speed and later rewired using a different circuit assembly methodology to exploit full-speed operation. A single small SoC often provides most of these electrical interface options in a form factor barely larger than a large postage stamp, available in the American hobby market (and elsewhere) for a few dollars, allowing fairly sophisticated breadboard projects to be created at modest expense.

6.12.5 Features:

- 2 Distribution Strips, 200 tie-points.
- 630 tie-points in IC/ circuit areas.
- ABS plastic with color legend.
- Dimension: 6.5*4.4*0.3 inch.
- Hole/Pitch Style: Square wire holes (2.54mm)
- ABS heat Distortion Temperature: 84° C (183° F)
- Rating: 300/3 to 5Amps.
- Insulation Resistance : 500MΩ / DC500V.

6.12.6 APPLICATION:

They are used for creating electrical connections between electronic components and single board computers or microcontrollers such as Arduino and Raspberry Pi.

6.13 LIGHT-EMITTING DIODE (LED):

A light-emitting diode (LED) is a semiconductor light source. LEDs are used as indicator lamps in many devices, and are increasingly used for lighting. When a light-emitting diode is forward biased (switched on), electrons are able to recombine with holes within the device, releasing energy in the form of photons.

This effect is called electroluminescence and the color of the light (corresponding to the energy of the photon) is determined by the energy gap of the semiconductor. An LED is often small in area (less than 1 mm²), and integrated optical components may be used to shape its radiation pattern. LEDs present many advantages over incandescent light sources including lower energy consumption, longer lifetime, improved robustness, smaller size, faster switching, and greater durability and reliability.



FIG (16) LIGHT EMITTING DIODE (LED)

Light-emitting diodes are used in applications as diverse as replacements for aviation lighting, automotive lighting as well as in traffic signals. The compact size, the possibility of narrow bandwidth, switching speed, and extreme reliability of LEDs has allowed new text and video displays and sensors to be developed, while their high switching rates are also useful in advanced communications technology.

6.13.1 Types of LED:

- Dimmer Switches. The LED is available with dimmer switches similar to other bulbs
- Color LED. ...
- LED Tubes for Lighting. ...
- SMD LED. ...
- COB LED. ...
- Graphene LED. ...
- Traditional and Inorganic LEDs. ...
- Maximum Brightness LEDs.

6.13.2 FEATURES:

- Light Generated by LED is Directional. A. LED is all forward directional lighting, not Omni as conventional light bulb. ...
- LED can Generate Different Light Color. A. ...
- Temperature will Affect LED Efficacy. A. ...
- Low Energy Consumption. A. ...
- Long Life.

6.13.3 APPLICATION:

- Aviation lighting
- fairy lights
- automotive headlamps
- advertising
- general lighting
- traffic signals
- camera flashes
- lighted wallpaper
- medical devices.

6.14 TRANSFORMER/ADAPTER:

6.14.1 TRANSFORMER:

Transformers are classified by their function, which is either step up or step down. Step-up transformers increase the voltage of the incoming current, while step-down transformers decrease the incoming current's voltage. The incoming voltage is referred to as the primary voltage, while the outgoing stream is referred to as secondary.

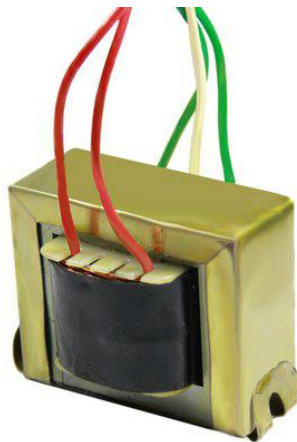
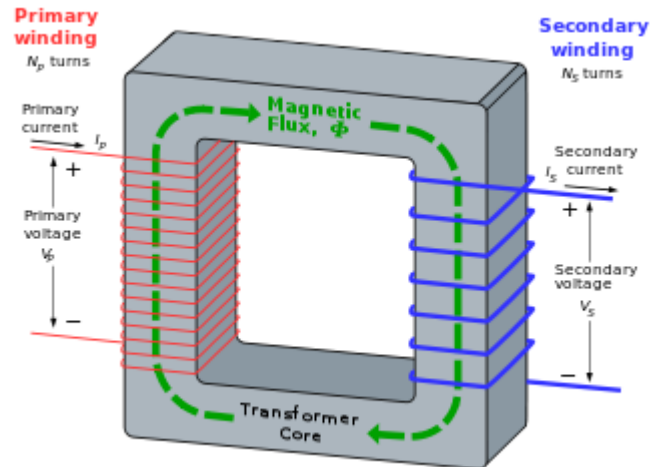


FIG (17) TRANSFORMER

Generally, step-up transformers are located at power generation plants, stepping up the voltage flowing from the power plant to long-distance distribution networks. Step-down transformers, on the other hand, decrease the voltage of power streams received at the local distribution level. The long-distance stream is first stepped down to a level acceptable for local distribution, and then stepped down again at each consumer node (residential homes and offices).

6.14.2 SPECIFICATIONS:

- Type Of Transformer: Step Down
- Input Voltage: 230V AC, 50 Hz
- Output Voltage: 9V AC
- Output Current: 1A
- Mounting: Vertical Mount Type
- Winding: Copper



6.14.3 Applications:

- It is used for increasing or decreasing the voltage in a circuit.
- It is used for electrically isolating two circuits.
- It is used in AC to DC rectifiers for reducing the High input AC voltages.
- It is used for impedance matching
- Current transformers are used for measurement purposes.
- Distribution transformers are used for reducing the voltage levels for our domestic appliances.
- Voltage stabilizer & regulators

6.14.4 ADAPTER:

An adapter is a physical device that allows one hardware or electronic interface to be adapted (accommodated without loss of function) to another hardware or electronic interface. In a computer, an adapter is often built into a [card](#) that can be inserted into a slot on the computer's motherboard. The card adapts information that is exchanged between the computer's microprocessor and the devices that the card supports.



FIG (18) ADAPTER

6.14.5 SPECIFICATION:

- Output Voltage: 19V DC.
- Output Current: 3.42A.
- Output Power: 65W.
- Input Voltage Range: 100V AC to 240V AC.
- Input Frequency: 50Hz to 60Hz.
- Connector: Inner diameter 2.5mm. External diameter 5.5mm. Depth 11.0mm.

6.15 PUSH BUTTON:

A push-button (also spelled pushbutton) or simply button is a simple switch mechanism for controlling some aspect of a machine or a process. Buttons are typically made out of hard material, usually plastic or metal. The surface is usually flat or shaped to accommodate the human finger or hand, so as to be easily depressed or pushed. Buttons are most often biased switches, though even many un-biased buttons (due to their physical nature) require a spring to return to their un-pushed state. Different people use different terms for the "pushing" of the button, such as press, depress, mash, and punch.

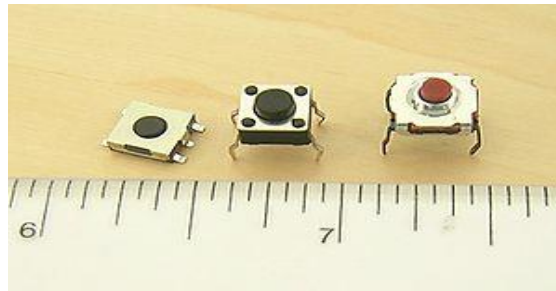


FIG (19) PUSH BUTTON

Uses:

In industrial and commercial applications push buttons can be linked together by a mechanical linkage so that the act of pushing one button causes the other button to be released. In this way, a stop button can "force" a start button to be released. This method of linkage is used in simple manual operations in which the machine or process have no electrical circuits for control.

Pushbuttons are often color-coded to associate them with their function so that the operator will not push the wrong button in error. Commonly used colors are red for stopping the machine or process and green for starting the machine or process.

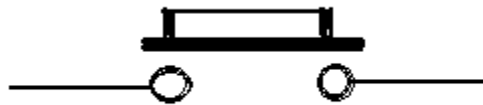
Red pushbuttons can also have large heads (mushroom shaped) for easy operation and to facilitate the stopping of a machine. These pushbuttons are called emergency stop buttons and are mandated by the electrical code in many jurisdictions for increased safety. This large mushroom shape can also be found in buttons for use with operators who need to wear gloves for their work and

could not actuate a regular flush-mounted push button. As an aid for operators and users in industrial or commercial applications, a pilot light is commonly added to draw the attention of the user and to provide feedback if the button is pushed. Typically this light is included into the center of the pushbutton and a lens replaces the pushbutton hard center disk.

The source of the energy to illuminate the light is not directly tied to the contacts on the back of the pushbutton but to the action the pushbutton controls. In this way a start button when pushed will cause the process or machine operation to be started and a secondary contact designed into the operation or process will close to turn on the pilot light and signify the action of pushing the button caused the resultant process or action to start.

In popular culture, the phrase "the button" refers to a (usually fictional) button that a military or government leader could press to launch nuclear weapons.

Push to ON button:



Initially the two contacts of the button are open. When the button is pressed they become connected. This makes the switching operation using the push button.

6.15.1 SPECIFICATION:

- Mode of Operation: Tactile feedback
- Power Rating: MAX 50mA 24V DC
- Insulation Resistance: 100Mohm at 100v
- Operating Force: 2.55 ± 0.69 N
- Contact Resistance: MAX 100mOhm
- Operating Temperature Range: -20 to +70 °C
- Storage Temperature Range: -20 to +70 °C

6.16 BUZZER:

A buzzer or beeper is an audio signaling device, which may be mechanical, electromechanical, or piezoelectric. Typical uses of buzzers and beepers include alarm devices, timers and confirmation of user input such as a mouse click or keystroke.



FIG (20) BUZZER

I) Mechanical:

A joy buzzer is an example of a purely mechanical buzzer.

II) Electro-mechanical:

Early devices were based on an electromechanical system identical to an electric bell without the metal gong. Similarly, a relay may be connected to interrupt its own actuating current, causing the contacts to buzz. Often these units were anchored to a wall or ceiling to use it as a sounding board. The word "buzzer" comes from the rasping noise that electromechanical buzzers made.

III) Piezoelectric:

A piezoelectric element may be driven by an oscillating electronic circuit or other audio signal source, driven with a piezoelectric audio amplifier. Sounds commonly used to indicate that a button has been pressed are a click, a ring or a beep.



FIG (21) PIEZOELECTRIC BUZZER

6.16.1 BUZZER INTERFACING:

If the proper potential is provided across the buzzer it generates the sound else not. It is of two pin configuration that is anode (+) and cathode (-). For interfacing to any device like controller etc. we directly connect the cathode to the GND and anode to the device like controller. Now when signal on anode goes high buzzer activates else does-not. Vice-versa is also true.

6.16.2 Features:

- Rated Voltage: 6V DC.
- Operating Voltage: 4-8V DC.
- Rated current: <30mA.
- Sound Type: Continuous **Beep**.
- Resonant Frequency: ~2300 Hz.
- Small and neat sealed package.
- Breadboard and Perf board friendly.

6.16.3 Applications of Buzzer:

- Alarming Circuits, where the user has to be alarmed about something
- Communication equipments
- Automobile electronics
- Portable equipments, due to its compact size

6.17 Integrated Circuit (IC):

An integrated circuit, or IC, is small chip that can function as an amplifier, oscillator, timer, microprocessor, or even computer memory. An IC is a small wafer, usually made of silicon, that can hold anywhere from hundreds to millions of transistors, resistors, and capacitors. These extremely small electronics can perform calculations and store data using either digital or analog technology.

Digital ICs use logic gates, which work only with values of ones and zeros. A low signal sent to a component on a digital IC will result in a value of 0, while a high signal creates a value of 1. Digital ICs are the kind you will usually find in computers, networking equipment, and most consumer electronics.

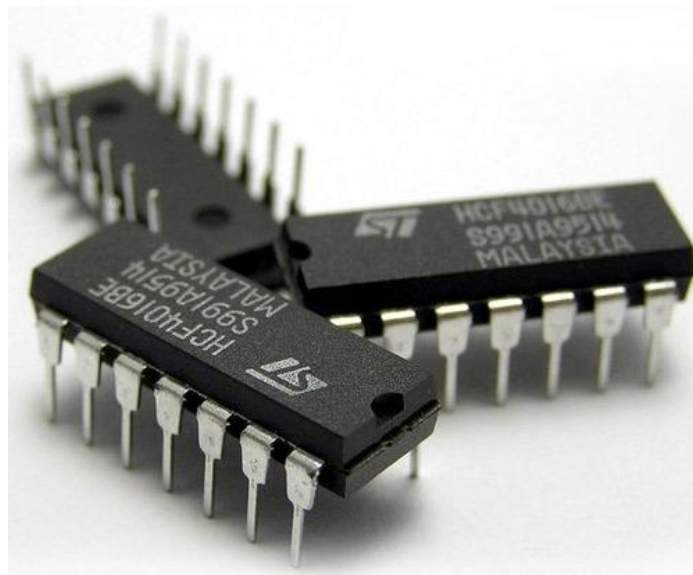
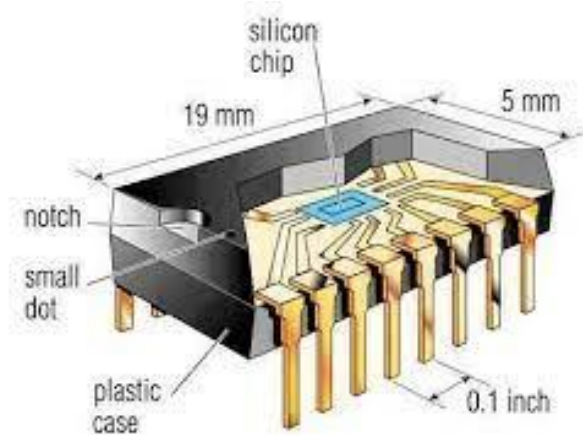


FIG (22) INTEGRATED CIRCUIT (IC)

Analog, or linear ICs work with continuous values. This means a component on a linear IC can take a value of any kind and output another value. The term "linear" is used since the output value is a linear function of the input. For example, a component on a linear IC may multiple an incoming value by a factor of 2.5 and output the result. Linear ICs are typically used in audio and radio frequency amplification.



6.17.1 SPECIFICATION:

- Input voltage range. “Input voltage range” refers to the range of input voltages in which the IC can operate. ...
- Output voltage range. ...
- Output current. ...
- Switching frequency. ...
- Operating temperature range.

6.17.2 APPLICATION:

Integrated circuits (IC) are being used in number of electronics application such as in field like:

- audio communication
- radio communication
- medical electronics
- instrumental control
- An important linear IC is operational amplifier (op-amp).

6.18 IC SOCKETS:

An IC socket can be defined as an electromechanical device, which provides a removable interface between the IC package and the system circuit board with minimal effect on signal integrity. A removable interface is the major reason for using a socket, and it is required for a variety of reasons, including ease of assembly, reworking, upgrading and cost savings. The cost advantage is saving the IC by not attaching it permanently to the PCB (printed circuit board). The socket is permanently (soldered) or semi-permanently (solder less) attached to the PCB, while the IC device can be inserted into or removed from the socket without disturbing the connections to the PCB. This allows the IC to function as it is soldered into the PCB, but it can be replaced by another IC or multiple ICs.



FIG (23) IC SOCKETS

The socket helps to test, evaluate and inspect the complete system. The socket also allows in the field for maintenance, testing, replacement or upgrades. This becomes a critical factor because of technology evolution. Sockets are a necessary component in designing today's electronic products. In the design process, sockets offer advantages in terms of flexibility of design, prototype evaluation, and product iteration; all geared toward speeding up time to market. During the design verification process and production startup, sockets provide

the ability to verify the consistency of the design in both the pre-production as well as in the ongoing production quality control process. Finally, in the end-use application, high-performance sockets may offer an attractive alternate to direct solder attach, especially in those applications where product upgradeability and field repair are important. Regardless of the application, both high performance and small footprint are desirable characteristics for the consideration of any socket.

6.18.1 CLASSIFIED INTO TWO CATEGORIES:

- Microprocessor, chipset, ASIC – IC development
- Mother board, various application boards – system development

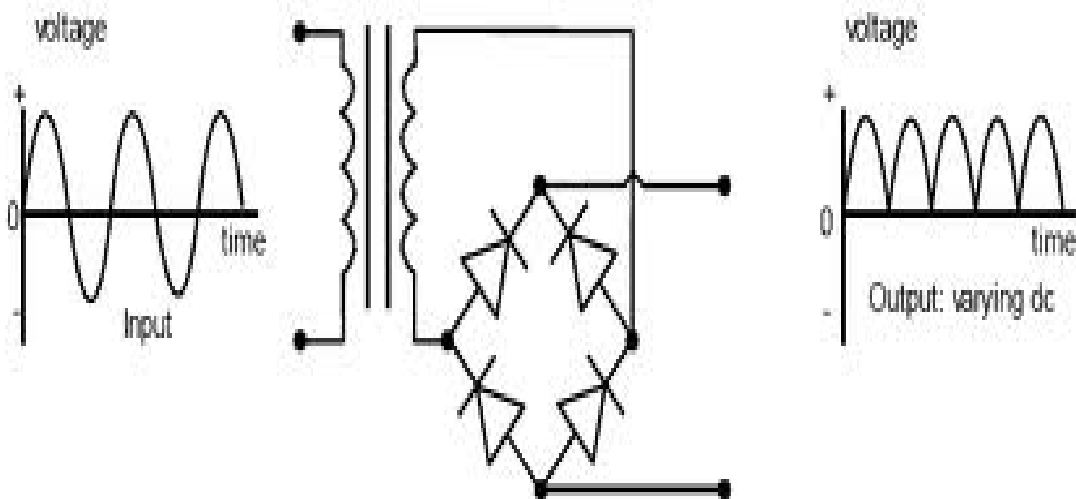
6.18.2 FOOTPRINT IS IMPORTANT WHENEVER A SOCKET IS CONSIDERED FOR:

- Design and development – small keep-out areas allow capacitors and resistors to be placed close to an IC; and similarity to end-use conditions
- Design verification – similarity to initial design and end-use conditions, signal integrity, etc.
- Burn-in application – Cost savings; cost per square inch of PCB utilization; speed-to-market
- Production – minimize board real estate and space for other components
- Failure analysis – similar to production board to weed out signal integrity and logic issues

6.19 RECTIFIER:

A rectifier is an electrical device that converts alternating current (AC), which periodically reverses direction, to direct current (DC), which flows in only one direction. The reverse operation is performed by the inverter.

The process is known as *rectification*, since it "straightens" the direction of current. Physically, rectifiers take a number of forms, including vacuum tube diodes, wet chemical cells, mercury-arc valves, stacks of copper and selenium oxide plates, semiconductor diodes, silicon-controlled rectifiers and other silicon-based semiconductor switches. Historically, even synchronous electromechanical switches and motors have been used. Early radio receivers, called crystal radios, used a "cat's whisker" of fine wire pressing on a crystal of galena (lead sulfide) to serve as a point-contact rectifier or "crystal detector".



Rectifiers have many uses, but are often found serving as components of DC power supplies and high-voltage direct current power transmission systems. Rectification may serve in roles other than to generate direct current for use as a source of power. As noted, detectors of radio signals serve as rectifiers. In gas heating systems flame rectification is used to detect presence of a flame.

Depending on the type of alternating current supply and the arrangement of the rectifier circuit, the output voltage may require additional smoothing to produce a uniform steady voltage. Many applications of rectifiers, such as power supplies for radio, television and computer equipment, require a *steady* constant DC voltage (as would be produced by a battery). In these applications the output of

the rectifier is smoothed by an electronic filter, which may be a capacitor, choke, or set of capacitors, chokes and resistors, possibly followed by a voltage regulator to produce a steady voltage.

6.19.1 TYPES OF RECTIFIERS:

- Single Phase & Three Phase Rectifiers.
- Half Wave & Full Wave Rectifiers.
- Bridge Rectifiers.
- Uncontrolled & Controlled Rectifiers.

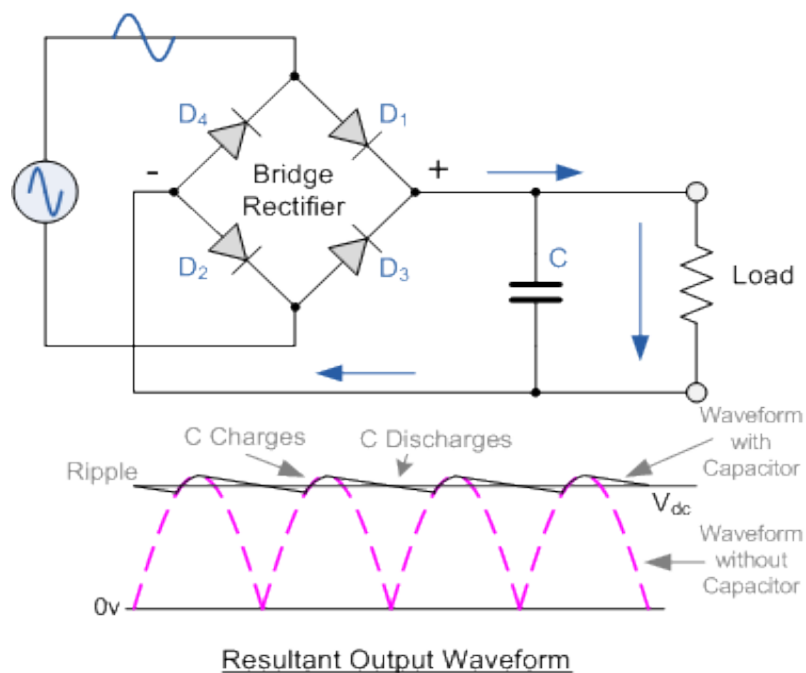
6.19.2 APPLICATIONS:

- A rectifier is used for powering appliances.
- These are used with transformers.
- Uses of rectifier while soldering.
- It is also used in AM radio.
- Uses of Rectifier in circuits.
- It is used for modulation.
- It is used in voltage multiplier.

6.20 FILTER:

Capacitive filter is used in this project. It removes the ripples from the output of rectifier and smoothens the D.C. Output received from this filter is constant until the mains voltage and load is maintained constant. However, if either of the two is varied, D.C. voltage received at this point changes. Therefore a regulator is applied at the output stage.

The simple capacitor filter is the most basic type of power supply filter. The use of this filter is very limited. It is sometimes used on extremely high-voltage, low-current power supplies for cathode-ray and similar electron tubes that require very little load current from the supply. This filter is also used in circuits where the power-supply ripple frequency is not critical and can be relatively high. Below figure can show how the capacitor charges and discharges.



6.20.1 TYPES OF FILTERS:

- low-pass filter
- high-pass filter
- band-pass filter
- notch/band-reject filter

6.20.2 ADVANTAGES OF FILTERS:

- They are economical or cost-effective
- Unlike passive filter circuits, Active Filter Circuits require power supply

6.20.3 DISADVANTAGES OF FILTERS:

- Circuits are bulky
- Limited Bandwidth
- Increased sensitivity to variation in circuit parameters

6.20.4 APPLICATIONS OF FILTERS:

- Filter Circuits are used to eliminate background Noise
- They are used in Radio tuning to a specific frequency
- Used in Pre-amplification, Equalization, Tone Control in Audio Systems
- They are also used in Signal Processing Circuits and Data Conversion
- Filter Circuits are extensively used in Medical Electronic Systems

7. SCHEMATIC DIAGRAM

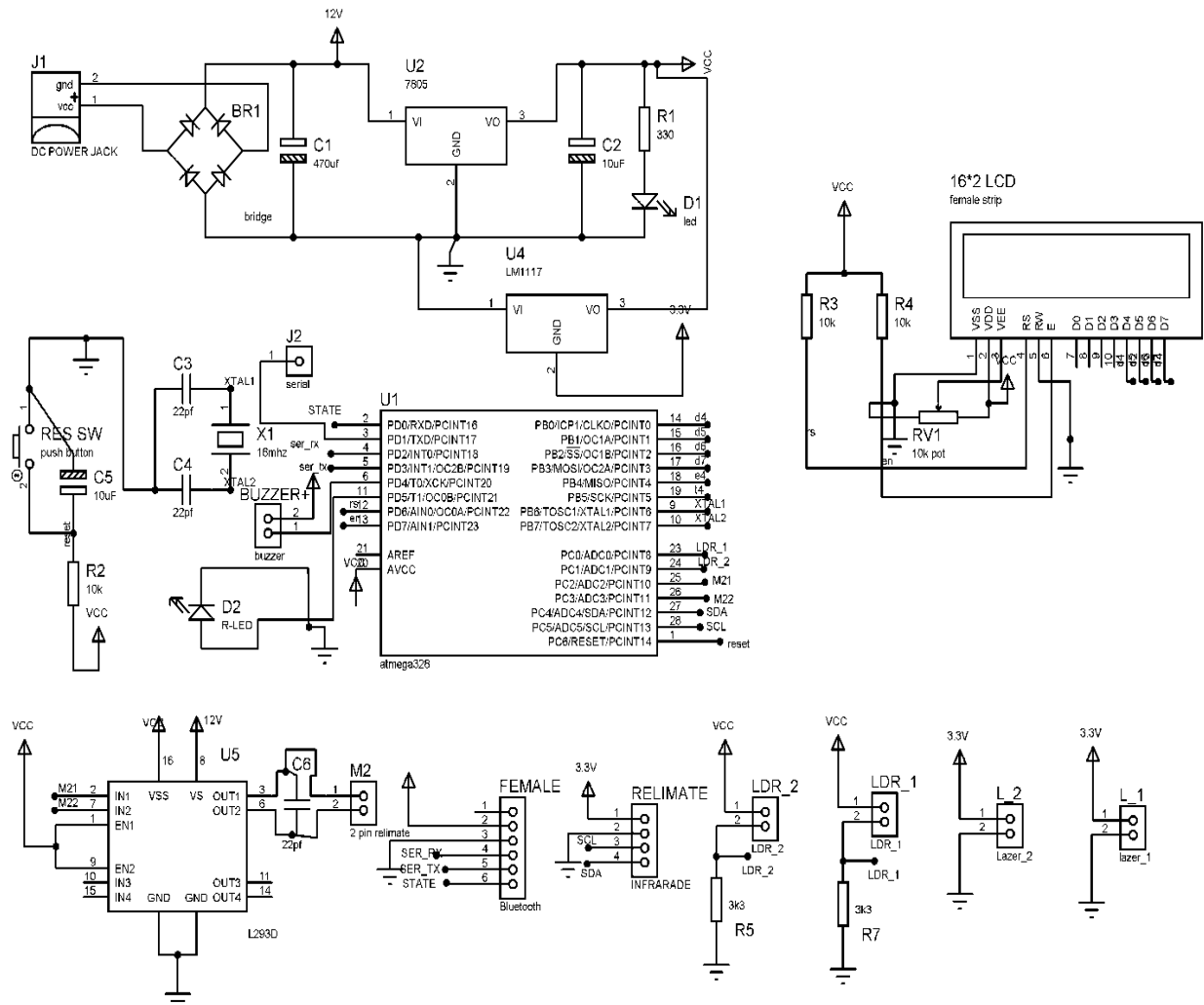


FIG (24) SCHEMATIC DIAGRAM

7.1 SCHEMATIC DIAGRAM DESCRIPTION

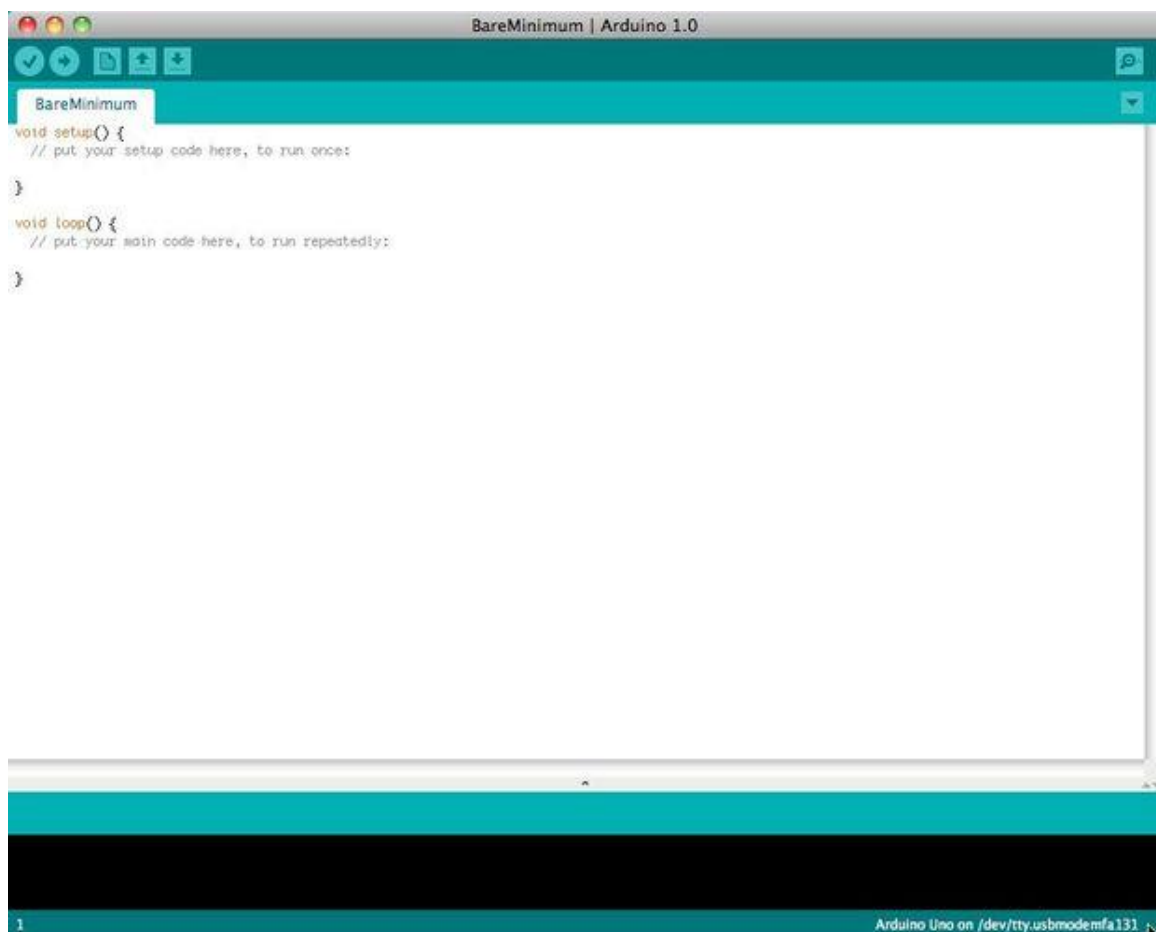
The circuit uses standard power supply comprising of a step-down transformer from 230V to 12V and 4 diodes forming a bridge rectifier that delivers pulsating dc which is then filtered by an electrolytic capacitor of about 470µF to 1000µF. The filtered dc being unregulated, IC LM7805 is used to get 5V DC constant at its pin no 3 irrespective of input DC varying from 7V to 15V. The input dc shall be varying in the event of input ac at 230volts section varies from 160V to 270V in the ratio of the transformer primary voltage V1 to secondary voltage V2 governed by the formula $V1/V2 = N1/N2$ i.e. no. of turns in the

primary to the no. of turns in the secondary remains unchanged V_2 is directly proportional to V_1 . Thus if the transformer delivers 12V at 220V input it will give 8.72V at 160V. Similarly at 270V it will give 14.72V. Thus the dc voltage at the input of the regulator changes from about 8V to 15V because of A.C voltage variation from 160V to 270V the regulator output will remain constant at 5V. The regulated 5V DC is further filtered by a small electrolytic capacitor of $10\mu\text{F}$ for any noise so generated by the circuit. One LED is connected of this 5V point in series with a current limiting resistor of 330Ω to the ground i.e., negative voltage to indicate 5V power supply availability. The unregulated 12V point is used for other applications as and when required.

8. SOFTWARE REQUIREMENTS

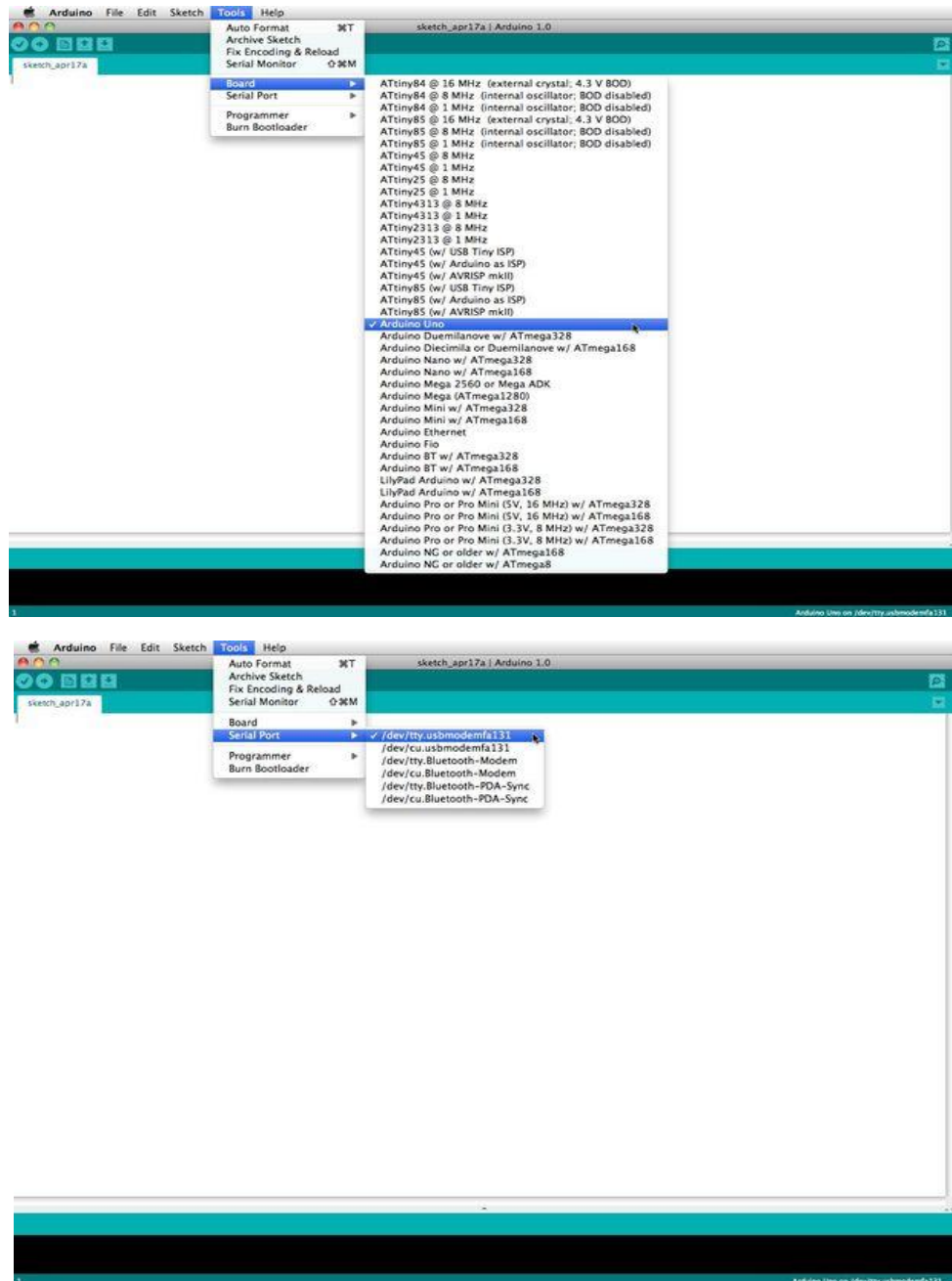
8.1 Arduino IDE:

IDE stands for “Integrated Development Environment” :it is an official software introduced by Arduino.cc, that is mainly used for editing, compiling and uploading the code in the Arduino Device. Almost all Arduino modules are compatible with this software that is an open source and is readily available to install and start compiling the code on the go. In this article, we will introduce the Software, how we can install it, and make it ready for developing applications using Arduino modules. we introduce some simulators for this interesting microcontrollers development tools. It is easy to use and learn.



Before you can start doing anything with the Arduino, you need to download and install the Arduino IDE (integrated development environment). From this point on we will be referring to the Arduino IDE as the Arduino Programmer. The Arduino Programmer is based on the Processing IDE and uses a variation of the C and C++ programming languages.

Settings:



Before you can start doing anything in the Arduino programmer, you must set the board-type and serial port.

To set the board, go to the following:

Tools --> Boards

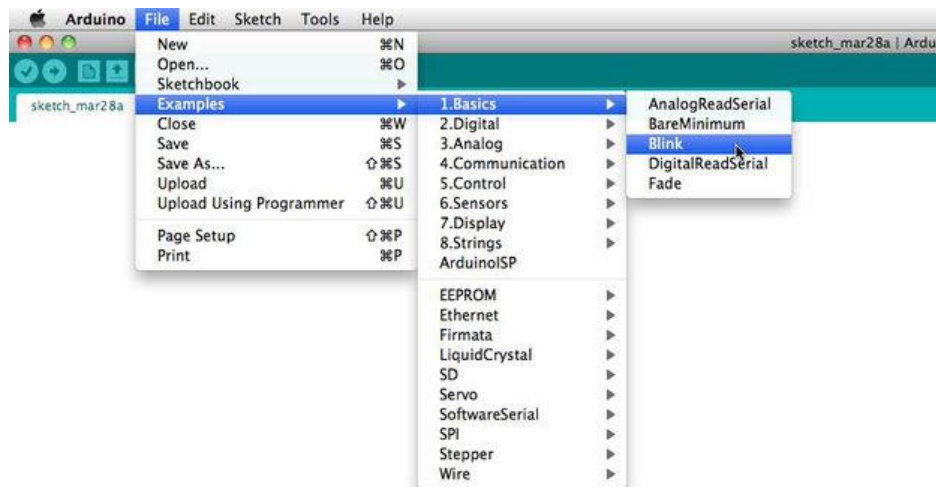
Select the version of board that you are using. Since I have an Arduino Uno plugged in, I obviously selected "Arduino Uno."

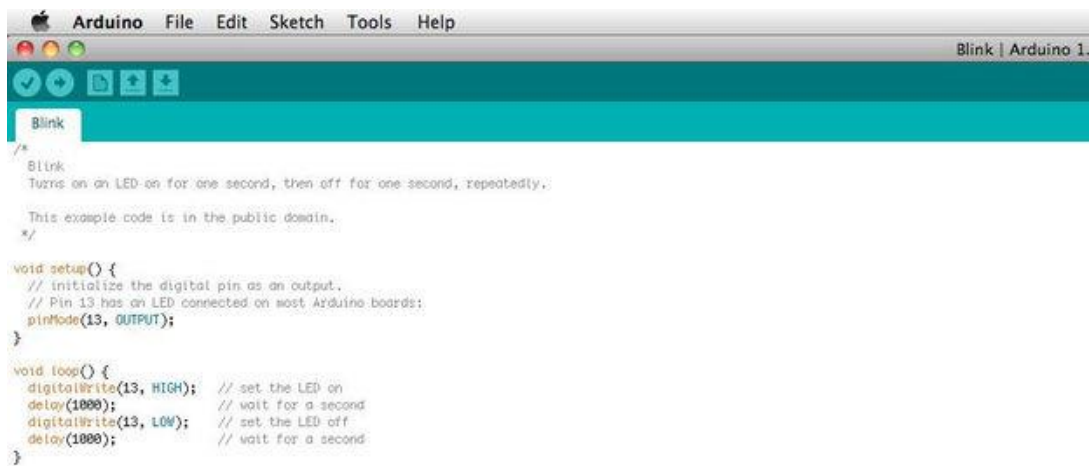
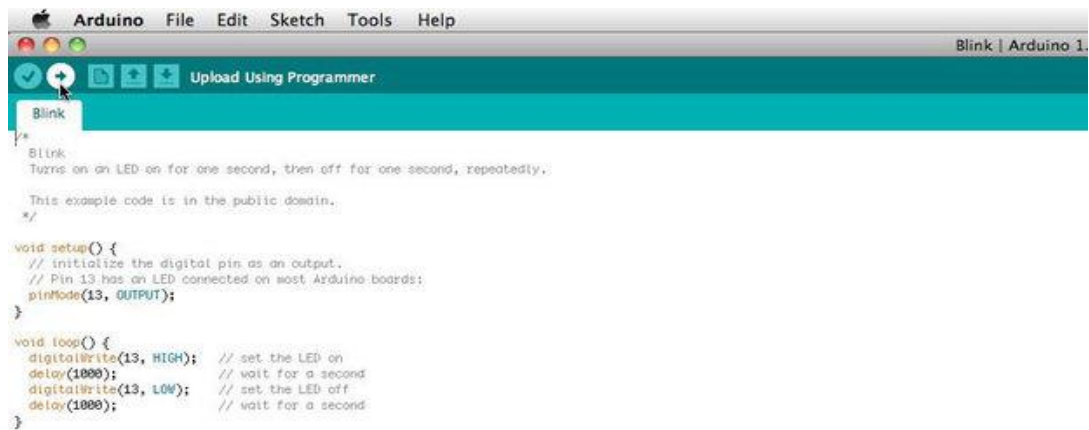
To set the serial port, go to the following:

Tools --> Serial Port

Select the serial port that looks like:

/dev/tty.usbmodem [random numbers]





Arduino programs are called sketches. The Arduino programmer comes with a ton of example sketches preloaded. This is great because even if you have never programmed anything in your life, you can load one of these sketches and get the Arduino to do something.

To get the LED tied to digital pin 13 to blink on and off, let's load the blink example.

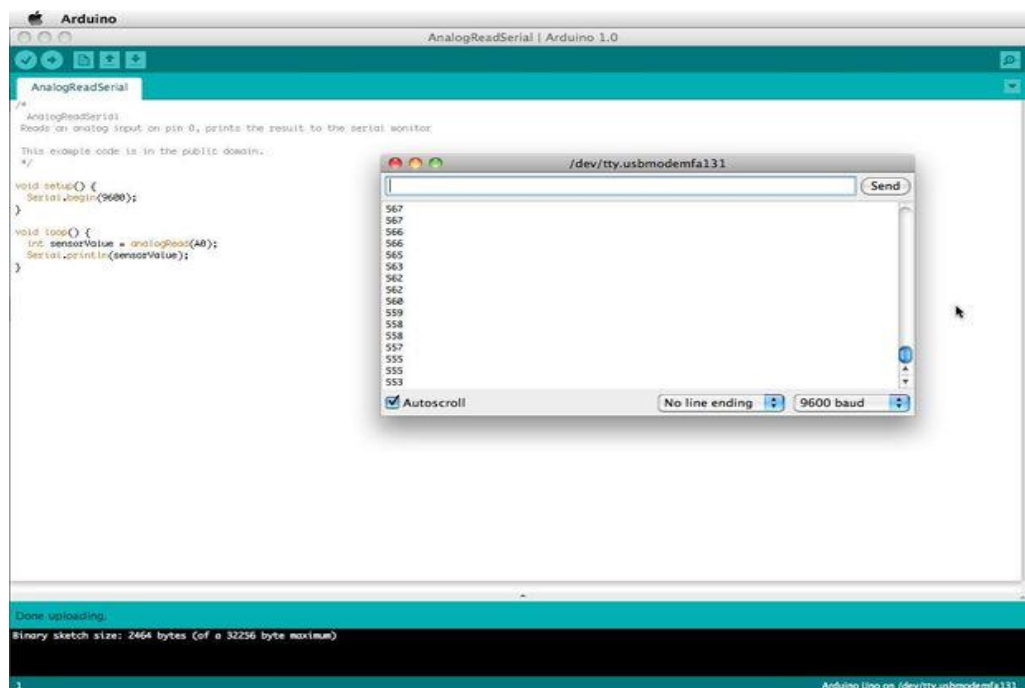
The blink example can be found here:
Files --> Examples --> Basics --> Blink

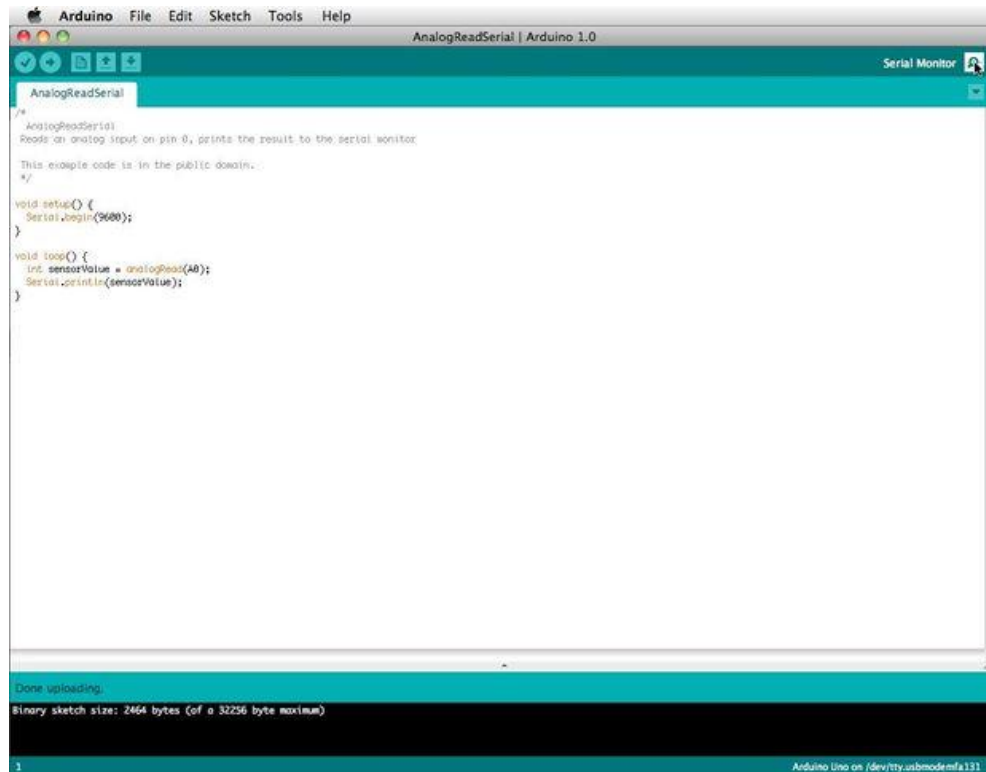
The blink example basically sets pin D13 as an output and then blinks the test LED on the Arduino board on and off every second.

Once the blink example is open, it can be installed onto the ATMEGA328 chip by pressing the upload button, which looks like an arrow pointing to the right.

Notice that the surface mount status LED connected to pin 13 on the Arduino will start to blink. You can change the rate of the blinking by changing the length of the delay and pressing the upload button again.

Serial monitor





The serial monitor allows your computer to connect serially with the Arduino. This is important because it takes data that your Arduino is receiving from sensors and other devices and displays it in real-time on your computer. Having this ability is invaluable to debug your code and understand what number values the chip is actually receiving.

For instance, connect center sweep (middle pin) of a potentiometer to A0, and the outer pins, respectively, to 5v and ground. Next upload the sketch shown below:

File --> Examples --> 1.Basics --> AnalogReadSerial

Click the button to engage the serial monitor which looks like a magnifying glass. You can now see the numbers being read by the analog pin in the serial monitor. When you turn the knob the numbers will increase and decrease.

The numbers will be between the range of 0 and 1023. The reason for this is that the analog pin is converting a voltage between 0 and 5V to a discreet number.

The Arduino has two different types of input pins, those being analog and digital.

To begin with, let's look at the digital input pins.

Digital input pins only have two possible states, which are on or off. These two on and off states are also referred to as:

- HIGH or LOW
- 1 or 0
- 5V or 0V.

This input is commonly used to sense the presence of voltage when a switch is opened or closed.

Digital inputs can also be used as the basis for countless digital communication protocols. By creating a 5V (HIGH) pulse or 0V (LOW) pulse, you can create a binary signal, the basis of all computing. This is useful for talking to digital sensors like a PING ultrasonic sensor, or communicating with other devices.

For a simple example of a digital input in use, connect a switch from digital pin 2 to 5V, a 10K resistor** from digital pin 2 to ground, and run the following code:

File --> Examples --> 2.Digital --> Button

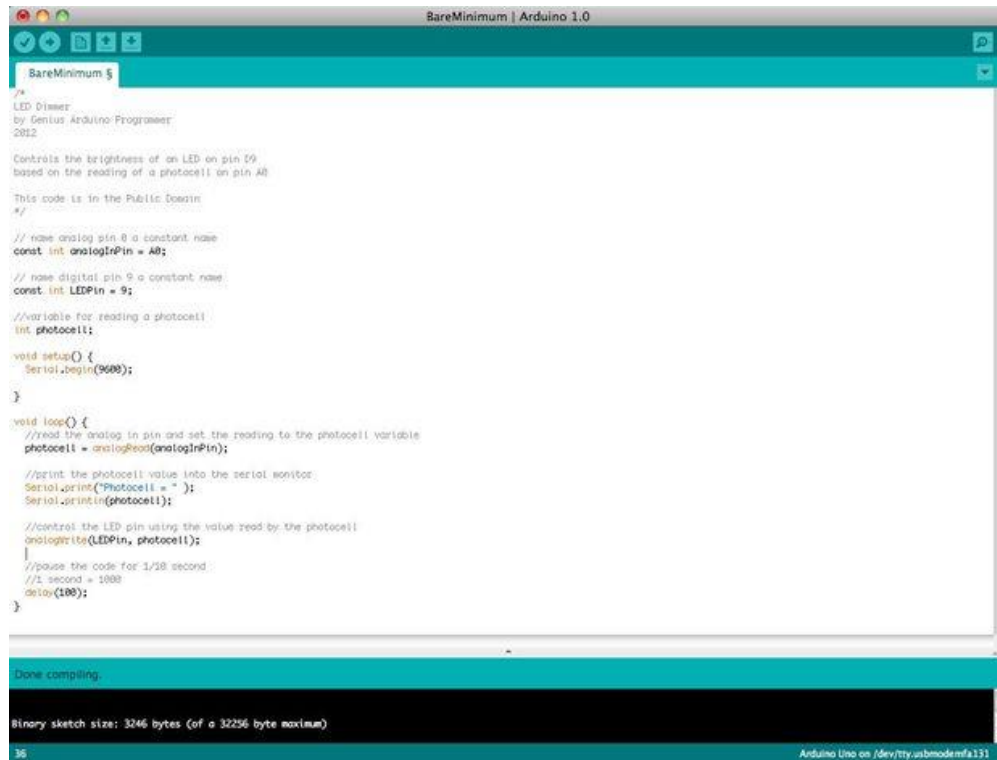
Analog in

Aside from the digital input pins, the Arduino also boasts a number of analog input pins.

Analog input pins take an analog signal and perform a 10-bit analog-to-digital (ADC) conversion to turn it into a number between 0 and 1023 (4.9mV steps).

This type of input is good for reading resistive sensors. These are basically sensors which provide resistance to the circuit. They are also good for reading a

varying voltage signal between 0 and 5V. This is useful when interfacing with various types of analog circuitry.

A screenshot of the Arduino IDE interface. The title bar reads 'BareMinimum | Arduino 1.0'. The main text area contains a C++ sketch for an LED dimmer. The sketch includes comments in Italian explaining its function: controlling an LED's brightness based on a photocell's reading. It defines two pins: analog pin 0 as 'analogInPin' and digital pin 9 as 'LEDpin'. The code includes a setup function to initialize serial communication at 9600 baud, and a loop function that reads the photocell value, prints it to the serial monitor, and controls the LED's brightness using a PWM signal. A delay of 1000 milliseconds is used between loop iterations. The status bar at the bottom indicates 'Done compiling.' and 'Binary sketch size: 3246 bytes (of a 32256 byte maximum)'. The bottom right corner shows 'Arduino Uno on /dev/tty.usbmodemfa131'.

To write your own code, you will need to learn some basic programming language syntax. In other words, you have to learn how to properly form the code for the programmer to understand it. You can think of this kind of like understanding grammar and punctuation. You can write an entire book without proper grammar and punctuation, but no one will be able to understand it, even if it is in English.

Some important things to keep in mind when writing your own code:

- An Arduino program is called a sketch.
- All code in an Arduino sketch is processed from top to bottom.
- Arduino sketches are typically broken into five parts.

1. The sketch usually starts with a header that explains what the sketch is doing, and who wrote it.

2. Next, it usually defines global variables. Often, this is where constant names are given to the different Arduino pins.
 3. After the initial variables are set, the Arduino begins the setup routine. In the setup function, we set initial conditions of variables when necessary, and run any preliminary code that we only want to run once. This is where serial communication is initiated, which is required for running the serial monitor.
 4. From the setup function, we go to the loop routine. This is the main routine of the sketch. This is not only where your main code goes, but it will be executed over and over, so long as the sketch continues to run.
 5. Below the loop routine, there is often other functions listed. These functions are user-defined and only activated when called in the setup and loop routine. When these functions are called, the Arduino processes all of the code in the function from top to bottom and then goes back to the next line in the sketch where it left off when the function was called. Functions are good because they allow you to run standard routines - over and over - without having to write the same lines of code over and over. You can simply call upon a function multiple times, and this will free up memory on the chip because the function routine is only written once. It also makes code easier to read. To learn how to form your own functions, check out this page.
- All of that said, the only two parts of the sketch which are mandatory are the Setup and Loop routines.
 - Code must be written in the Arduino Language, which is roughly based on C.
 - Almost all statements written in the Arduino language must end with a ;
 - Conditionals (such as if statements and for loops) do not need a ;
 - Conditionals have their own rules and can be found under "Control Structures" on the Arduino Language page
 - Variables are storage compartments for numbers. You can pass values into and out of variables. Variables must be defined (stated in the code) before they can be used and need to have a data type associated with it. To learn some of the basic data types, review the Language Page.

Okay! So let us say we want to write code that reads a photocell connected to pin A0, and use the reading we get from the photocell to control the brightness of an LED connected to pin D9.

First, we want to open the BareMinimum sketch, which can be found at:
File --> Examples --> 1.Basic --> BareMinimum

The BareMinimum Sketch should look like this:

```
<pre>void setup() {  
  // put your setup code here, to run once:  
  
}  
  
void loop() {  
  // put your main code here, to run repeatedly:  
  
}
```

Next, lets put a header on the code, so other people know about what we are making, why, and under what terms:

```
<pre>/*  
LED Dimmer  
by Genius Arduino Programmer  
2012  
  
Controls the brightness of an LED on pin D9  
based on the reading of a photocell on pin A0
```


This code is in the Public Domain

```
*/
```

```
void setup() {
```

```
  // put your setup code here, to run once:
```

```
}
```

```
void loop() {
```

```
  // put your main code here, to run repeatedly:
```

```
}
```

Once that is all squared away, let us define the pin names, and establish variables:

```
<pre>/*
```

```
LED Dimmer
```

```
by Genius Arduino Programmer
```

```
2012
```

```
Controls the brightness of an LED on pin D9  
based on the reading of a photocell on pin A0
```

```
This code is in the Public Domain
```

```
*/
```

```
// name analog pin 0 a constant name
const int analogInPin = A0;

// name digital pin 9 a constant name
const int LEDPin = 9;

//variable for reading a photocell
int photocell;

void setup() {
  // put your setup code here, to run once:

}

void loop() {
  // put your main code here, to run repeatedly:

}
```

Now that variables and pin names are set, let us write the actual code:

```
<pre>/*
LED Dimmer
by Genius Arduino Programmer
2012
```

Controls the brightness of an LED on pin D9
based on the reading of a photocell on pin A0

This code is in the Public Domain

```
*/  
  
// name analog pin 0 a constant name  
const int analogInPin = A0;  
  
// name digital pin 9 a constant name  
const int LEDPin = 9;  
  
//variable for reading a photocell  
int photocell;  
  
void setup() {  
  //nothing here right now  
  
}  
  
void loop() {  
  //read the analog in pin and set the reading to the photocell variable  
  photocell = analogRead(analogInPin);  
  
  //control the LED pin using the value read by the photocell  
  analogWrite(LEDPin, photocell);
```

```
//pause the code for 1/10 second
//1 second = 1000
delay(100);
}
```

If we want to see what numbers the analog pin is actually reading from the photocell, we will need to use the serial monitor. Let's activate the serial port and output those numbers:

```
<pre>/*
LED Dimmer
by Genius Arduino Programmer
2012

Controls the brightness of an LED on pin D9
based on the reading of a photocell on pin A0

This code is in the Public Domain
*/

// name analog pin 0 a constant name
const int analogInPin = A0;

// name digital pin 9 a constant name
const int LEDPin = 9;

//variable for reading a photocell
```

```
int photocell;

void setup() {
  Serial.begin(9600);

}

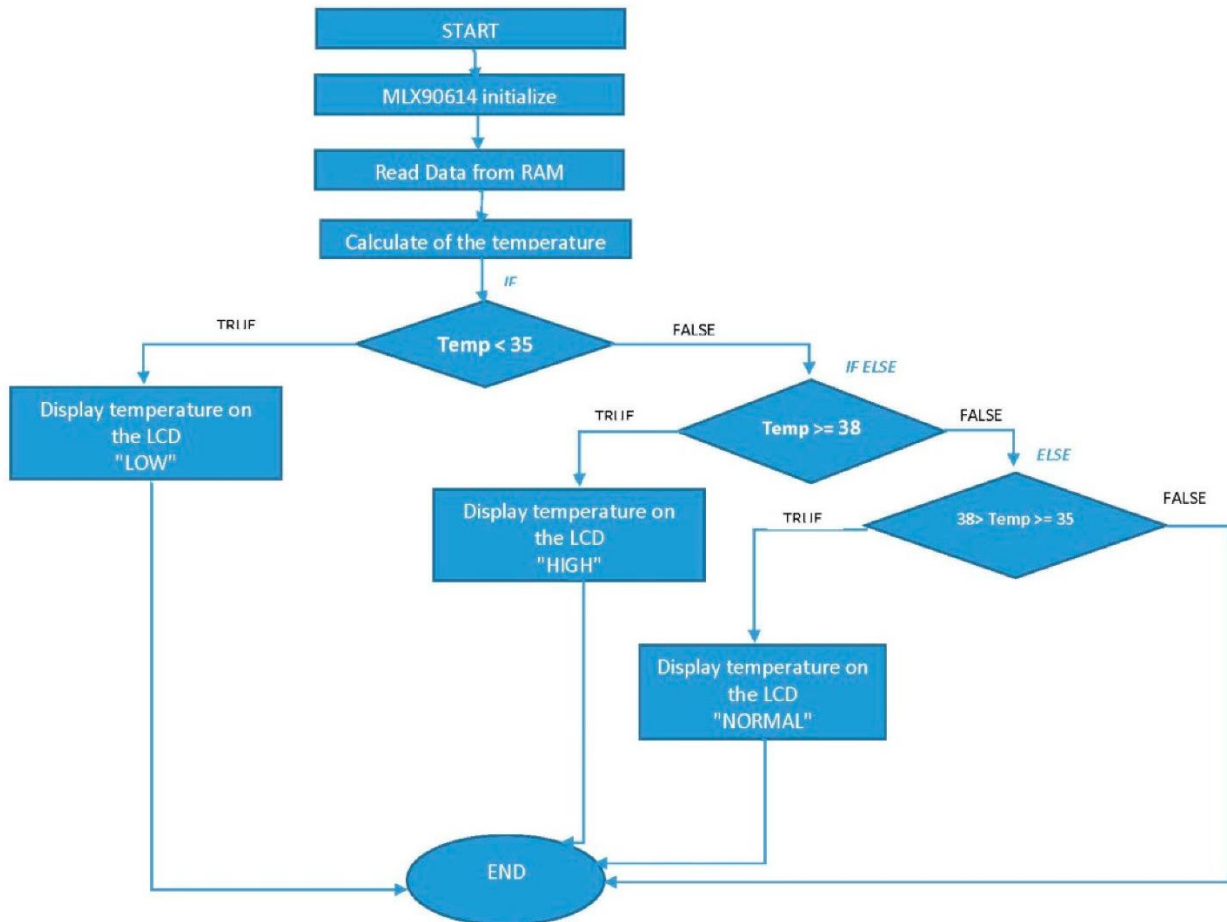
void loop() {
  //read the analog in pin and set the reading to the photocell variable
  photocell = analogRead(analogInPin);

  //print the photocell value into the serial monitor
  Serial.print("Photocell = " );
  Serial.println(photocell);

  //control the LED pin using the value read by the photocell
  analogWrite(LEDPin, photocell);

  //pause the code for 1/10 second
  //1 second = 1000
  delay(100);
}
```

9. FLOW CHART



the steps for the configuration and programming of the MLX90614 temperature sensor on Arduino software are reported. In particular, a normal value is assumed if the measured temperature is in a range between 35 °C and 38 °C then door/cabin will open up; for a temperature value greater than 38 °C, an alarm (BUZZER) should sound up and the door/cabin doesn't open, while for measured values lower than 35 °C, the temperature is assumed to be low. These results are displayed on the LED screen to be viewed by the user.

10. PROGRAM

```
#include <LiquidCrystal.h>
#include <iotgecko.h>
//#include <SoftwareSerial.h>
//SoftwareSerial mySerial(3, 2); // RX, TX
const int Add1 = 6;
const int Add0 = 7; //1st
const int data0 = 13;
const int data1 = 12;
const int data2 = 11;
const int data3 = 10;
const int VT = 8;
int device1 = 0;
int device2 = 0;
int device3 = 0;
int device4 = 0;
int x =0;
long t1=0;
#define esp_baudrate 115200
iotgecko gecko = iotgecko(esp_baudrate);

bool notConected = true;
bool login = false;
```

```

String id = "thomasikenna@gmail.com";//iotgecko login id
String pass = "8344";//iotgecko login password
String ssid = "iot";//Wifi SSID
String pass_key = "project1234";//Wifi password


const int no_of_data = 5;
String patient_data[no_of_data];
LiquidCrystal lcd(A0, A1, A2, A3, A4, A5); //RS, EN, D4, D5, D6, D7
void setup() {
    // put your setup code here, to run once:
    Serial.begin(115200);
    Serial.println(F("RF based hospital asset"));
    lcd.begin(16, 2);
    Serial.begin(9600);
    pinMode(Add1,OUTPUT);
    pinMode(Add0, OUTPUT );
    pinMode(VT, INPUT );
    pinMode(data0, INPUT );
    pinMode(data1, INPUT );
    pinMode(data2, INPUT );
    pinMode(data3, INPUT );
    delay(500);
    lcd.clear();
    lcd.setCursor(1,0);
    lcd.print("Connecting to");

```



```

lcd.setCursor(4,1);
lcd.print(" Wifi");
delay(1000);
Serial.println(F("connnecting to wifi"));
Serial.print(ssid);
Serial.print("\t");
Serial.println(pass_key);
while(notConected)
{
    if(gecko.GeckoConnect(ssid,pass_key))//connect to wifi with given SSID and
password
    {
        Serial.println(F("conncted to wifi..."));
        lcd.clear();
        lcd.setCursor(0,0);
        lcd.print("Wifi connected");
        delay(2000);
        notConected = false;
    }
    else
    {
        Serial.println(F("can't connect to wifi"));
        lcd.clear();
        lcd.setCursor(0,0);
        lcd.print("Can't Connect");
        lcd.setCursor(1,1);
    }
}

```

```

    lcd.print("to Wifi");
    delay(2000);
}
delay(1000);
}
Serial.println(F("connecting to iotgecko.com"));
while(!login)
{
    if(gecko.GeckoVerify(id,pass))//login to iotgecko.com with given ID and
password
    {
        Serial.println(F("connected succesfully"));
        lcd.clear();
        lcd.setCursor(0,0);
        lcd.print("Connected to");
        lcd.setCursor(1,1);
        lcd.print(" iotgecko");
        delay(2000);
        login = true;
    }
else
{
    Serial.println(F("fail to connect"));
    lcd.clear();
    lcd.setCursor(0,0);
    lcd.print("connection to");

```

```

        lcd.setCursor(0,1);
        lcd.print("iotgecko failed");
        delay(2000);
    }
}

}

void loop()
{
    // put your main code here, to run repeatedly:
    lcd.clear();
    lcd.setCursor(3,0);
    lcd.print("RF hospital");
    lcd.setCursor(5,1);
    lcd.print("Asset");
    delay(2000); // delay 2 secs
    while(1)
    {
        scan();
        patient_data[0] = (String)1; // room 1
        patient_data[1] = (String)device1;
        patient_data[2] = (String)device2;
        patient_data[3] = (String)device3;
    }
}

```

```

patient_data[4] = (String)device4;
lcd.clear();
if (device1==1)
{
    lcd.setCursor(0,0); // row, coloumn
    lcd.print("Device1");
}
if (device2==1)
{
    lcd.setCursor(9,0); // row, coloumn
    lcd.print("Device2");
}
if (device3==1)
{
    lcd.setCursor(0,1); // row, coloumn
    lcd.print("Device3");
}
if (device4==1)
{
    lcd.setCursor(9,1); // row, coloumn
    lcd.print("Device4");
}
if((device1!=1) && (device2!=1) && (device3!=1) && (device4!=1))
{
    lcd.clear();

```

```

    lcd.print("No devices");
    lcd.setCursor(0,1);
    lcd.print("found..");
}

int iot_status = gecko.SendGParams(patient_data,no_of_data);//send data to
iotgecko.com

if(iot_status == InvalidUserIdOrPassword)
{
    Serial.println(F("Invalid UserId or Password"));
    while(1);
}
else if(iot_status == InvalidData)
{
    Serial.println(F("Invalid Data"));
    while(1);
}
else if(iot_status == HitLimitExceed)
{
    Serial.println(F("Limit Exceed"));
}
else if(iot_status == VALID)
{
    Serial.println(F("data send succesfully....:"));
}
else
{

```

```

Serial.println(F("connection lost"));
while(!gecko.GeckoReconnect())//reconnect to the iotgecko.com
{
    Serial.println(F("connction failed.....reconnecting"));
    delay(2000);
}
Serial.println(F("connected succesfully"));
delay(1000);
}
}
}

```

```

void scan()
{
    lcd.clear();
    lcd.setCursor(2,0);
    lcd.print("Scanning for");
    lcd.setCursor(4,1);
    lcd.print("Devices.");
    digitalWrite(Add0, LOW);
    digitalWrite(Add1, LOW);
    t1 = millis();
    while(1)
    {

```

```

    if
    ((digitalRead(VT)==HIGH)&&(digitalRead(data0)==HIGH)&&(digitalRead(data1)==LOW)&&(digitalRead(data2)==LOW)&&(digitalRead(data3)==LOW))
    {
        device1 = 1;
//    lcd.clear();
//    lcd.print("Device1");
        break;
    }
    if((millis() - t1) >= 8000)
    {
        device1 = 0;
        break;
    }
}
lcd.print(".");
digitalWrite(Add0, LOW);
digitalWrite(Add1, HIGH);
t1 = millis();
while(1)
{

if((digitalRead(VT)==HIGH)&&(digitalRead(data0)==LOW)&&(digitalRead(data1)==HIGH)&&(digitalRead(data2)==LOW)&&(digitalRead(data3)==LOW))
{
    device2 = 1;
//    lcd.clear();

```

```

//    lcd.print("Device2");
    break;
}
if((millis() - t1) >= 8000)
{
    device2 = 0;
    break;
}
}
lcd.print(".");
digitalWrite(Add0, HIGH);
digitalWrite(Add1, LOW);
t1 = millis();
while(1)
{
    if
((digitalRead(VT)==HIGH)&&(digitalRead(data0)==LOW)&&(digitalRead(data1)==LOW)&&(digitalRead(data2)==HIGH)&&(digitalRead(data3)==LOW))
    {
        device3 = 1;
//    lcd.clear();
//    lcd.print("Device3");
        break;
    }
    if((millis() - t1) >= 8000)
    {

```



```

        device3 = 0;
        break;
    }
}
lcd.print(".");
digitalWrite(Add0, HIGH);
digitalWrite(Add1, HIGH);
t1 = millis();
while(1)
{
    if
((digitalRead(VT)==HIGH)&&(digitalRead(data1)==LOW)&&(digitalRead(data0)==LOW)&&(digitalRead(data2)==LOW)&&(digitalRead(data3)==HIGH))
    {
        device4 = 1;
//    lcd.clear();
//    lcd.print("Device4");
        break;
    }
    if((millis() - t1) >= 8000)
    {
        device4 = 0;
        break;
    }
}
if((device1!=1) && (device2!=1) && (device3!=1) && (device4!=1))

```

```
{  
  // patient_data[1] = 0 ;  
  // patient_data[2] = 0 ;  
  // patient_data[3] = 0 ;  
  // patient_data[4] = 0 ;  
  lcd.clear();  
  lcd.print("No devices");  
  lcd.setCursor(0,1);  
  lcd.print("found..");  
}  
delay(3000);  
}
```

11. ADVANTAGES

- Auto temperature detector is Easy to install and easy to test.
- Auto temperature detector for entrances for covid safety can check temperature without any contact (Zero Contact)
- It have the ability of Faster Testing without any disturbance.
- Due to fast testing there is No Long Queues forms
- There is No Errors due to manual work

12. DISADVANTAGES

- In this circuit there is lots of component that Requires Power Supply.

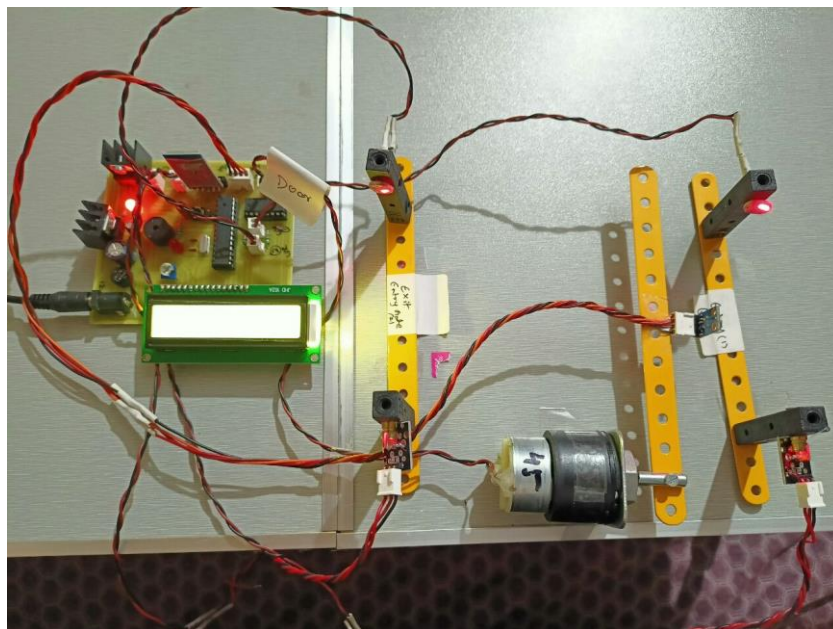
13. FUTURE SCOPE

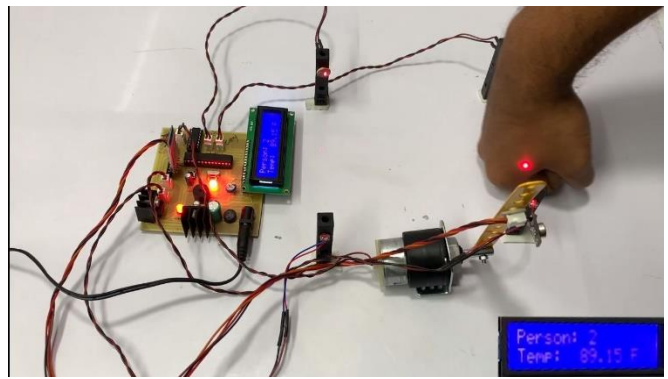
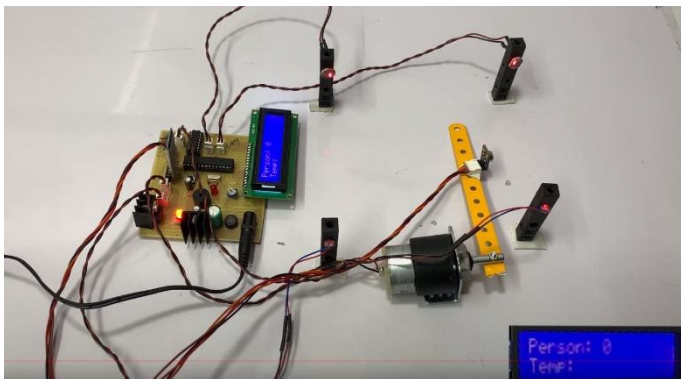
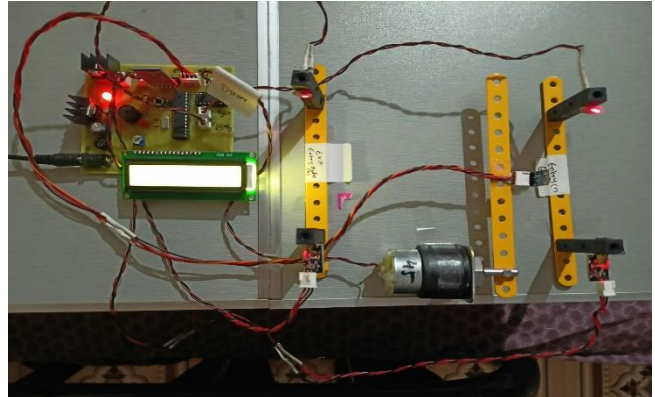
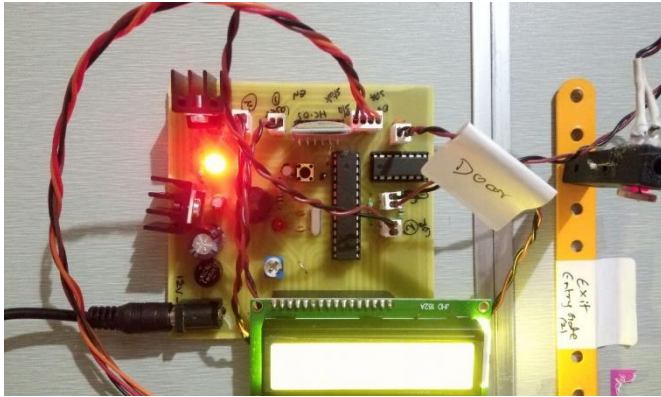
- Auto temperature detector for entrances can be used in public transportation areas like Bus stations, Railways stations and also at Airports
- Auto temperature detector finds its main application in educational premises like schools, colleges, university campuses.
- It's also helpful at industrial areas like in workshop, office entrances, etc.
- In shopping mall, Hotel & Restaurant Entrances we can use this project with zero contact.

14. RESULT

If a person passing through door/gate, auto temperature detector
Detect the temperature of that person ,if the temperature is in normal
form i.e under 99°F (37.2°C). then the door/gate will open automatically.
if the temperature of that person is above 99°F (37.2°C) then it will not
open the door/gate and it will display that the person have high temperature. And
the auto temperature detector for entrance for covid safety have one more feature
that we can control how many person are allowed in that room/cabin ,after
detecting the temperature it will count the person going through it but if we set
the limit of 10 person allowed in a room/cabin then the 11th person can't go in the
room/cabin. it will display that room is full.

BELOW ARE SOME IMAGE OF ASSEMBLY:





VIDEO OF PROJECT EXECUTION:

<https://drive.google.com/file/d/1G0aNozdtxG3R4aAlBDrYFyudmWDwJ6Ug/view?usp=sharing>

OR

<https://vimeo.com/566822384>



VID20210621091137.
mp4

15. CONCLUSION

An integrated platform combining multiple sensors for the simultaneous non-contact measurement of relative humidity, ambient temperature, and body temperature has been presented in this work. The operating principle and the full hardware design has been accurately described. Numerical simulations, reporting the curves for the three assumed variable parameters are shown and discussed to provide evidence of the nominal design. Experimental validations have also been provided to demonstrate the improved accuracy of infrared temperature measurements achieved with the proposed approach. This is particularly useful for detecting fevers and therefore possible infections of COVID-19 and other illnesses.

17. BIBLIOGRAPHY

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17.2 WEBSITES

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